

EMI Seminar 2017



KTH

21 February
By Thomas Ginell

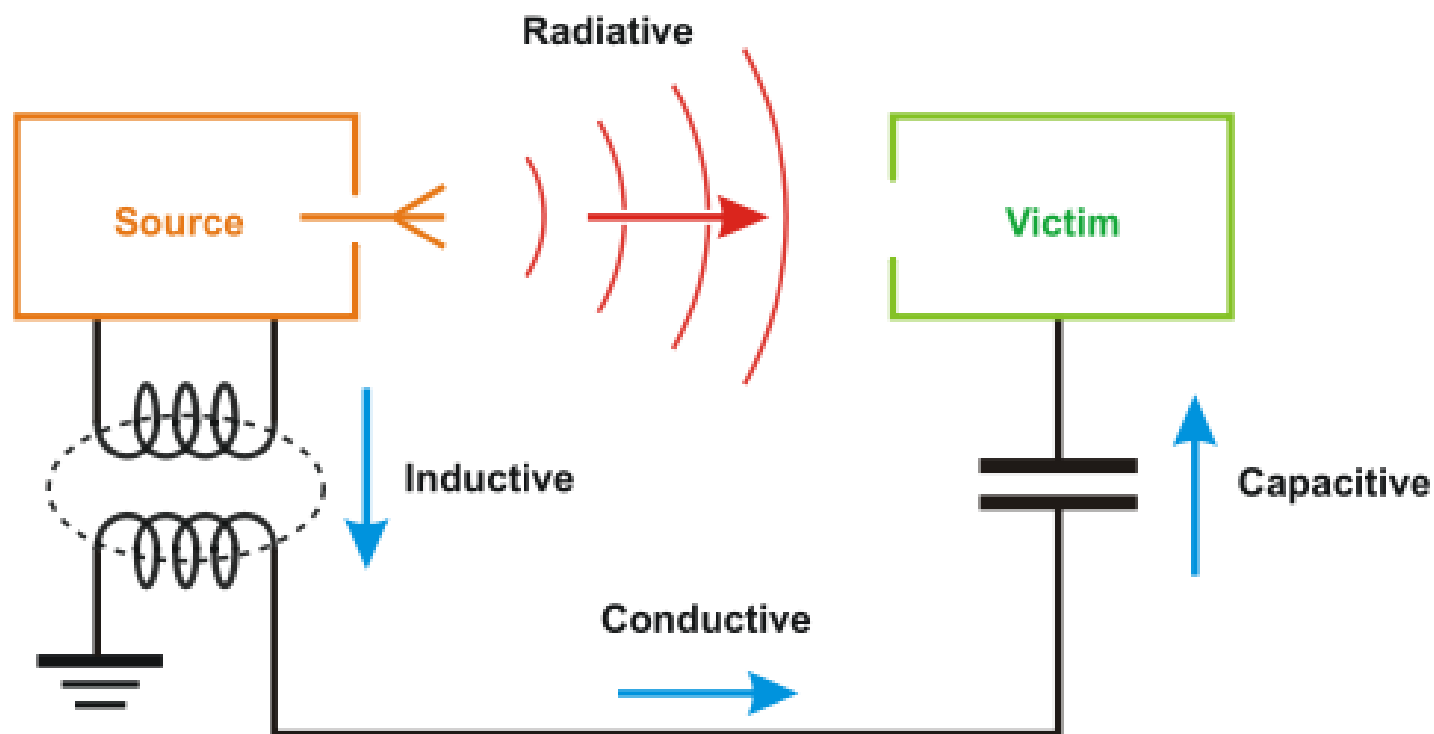
ANALOG EXCELLENCE

INNOVATION LED RECOVERY



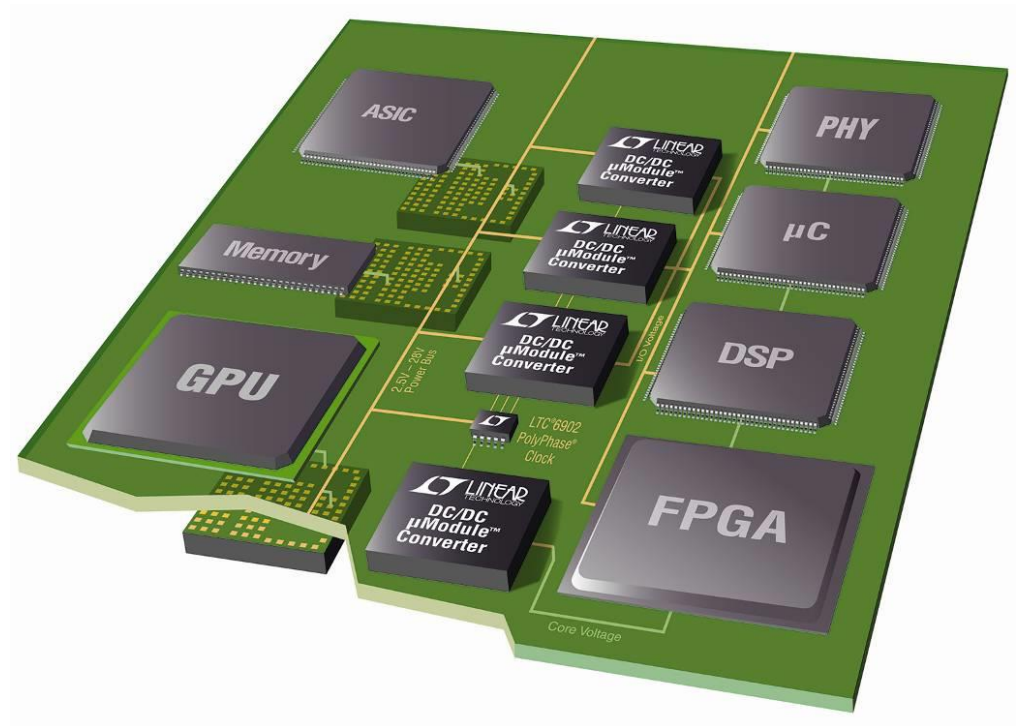
- Basic switch mode power design
- Circuit EMI, ripple, noise, component behavior
- Basic electromagnetics

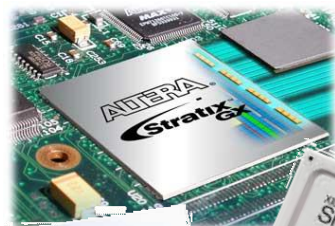
Electromagnetic Interference



Hardware
Needs

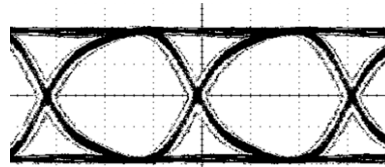
POWER



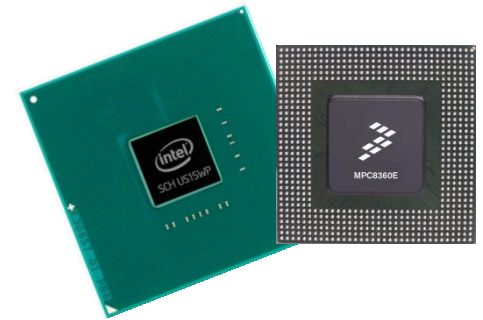
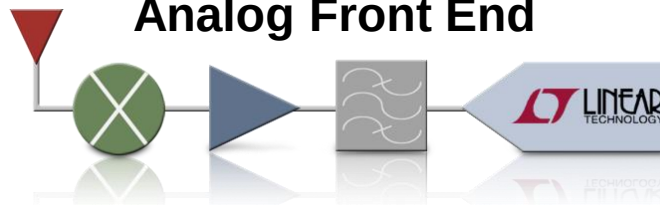


FPGA

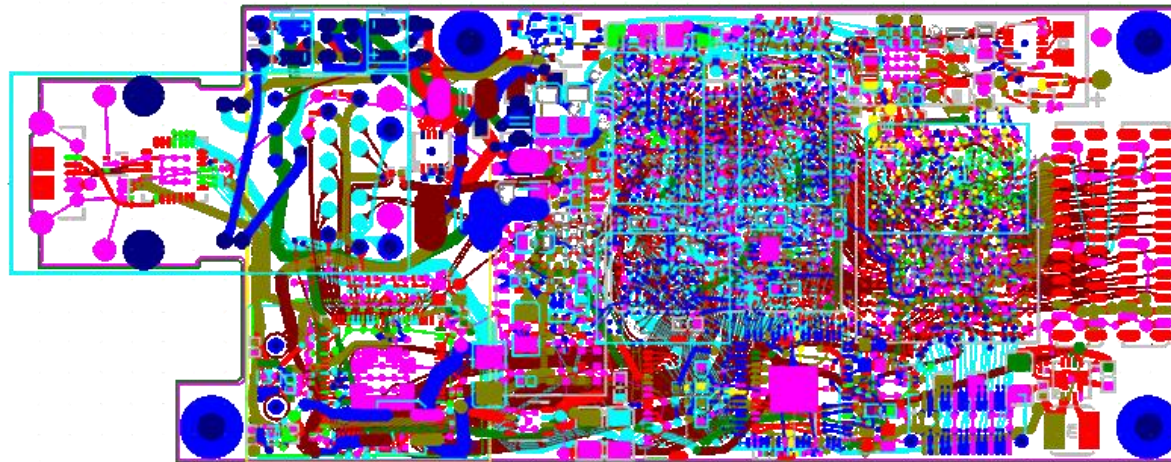
High Speed Data Interface



Analog Front End

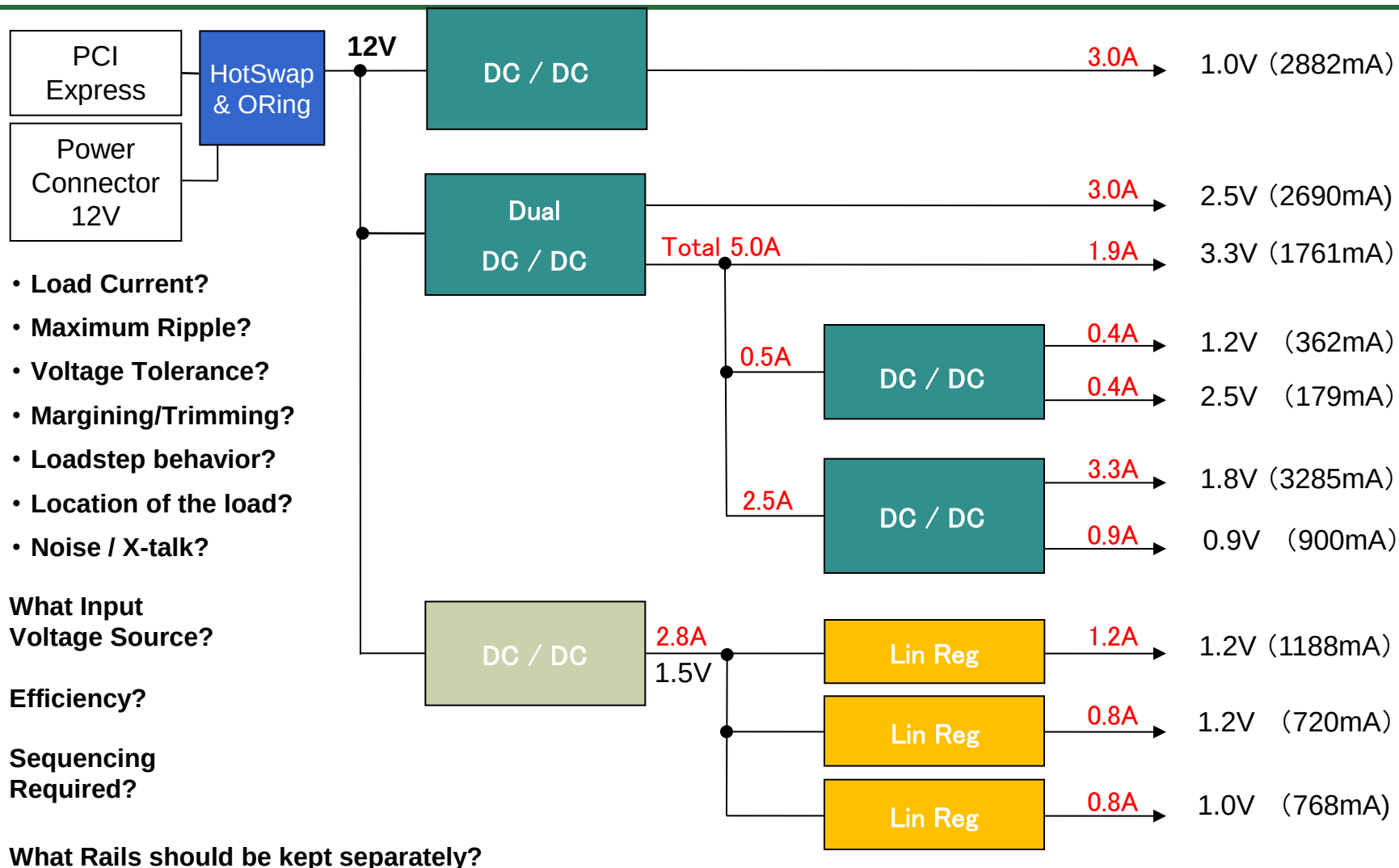


**Processor
DSP**



And everything has to fit in here !

Detailed Voltage Rail Map

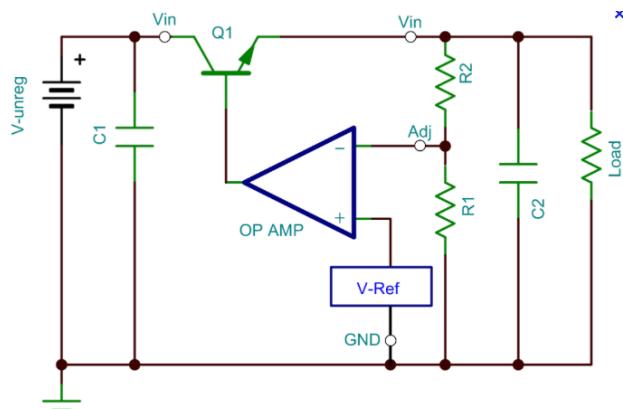


Which Part to Choose?

Linear Regulator

- Easy to design and use
- No magnetics
- Low noise
- **Low efficiency**

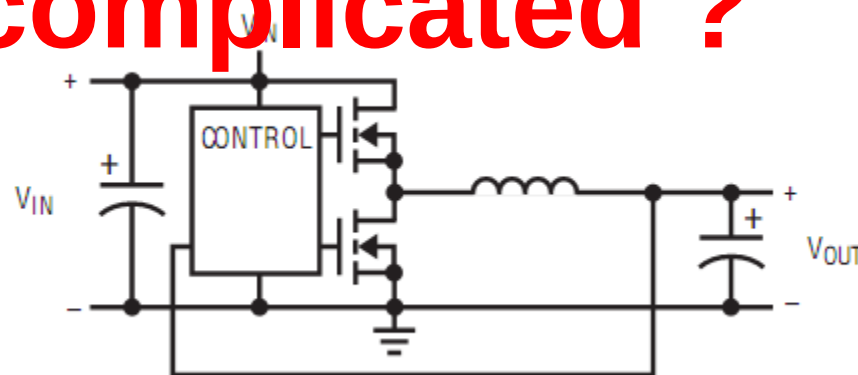
Heat Power
consumption



Switching Regulator

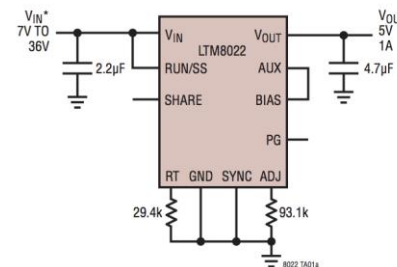
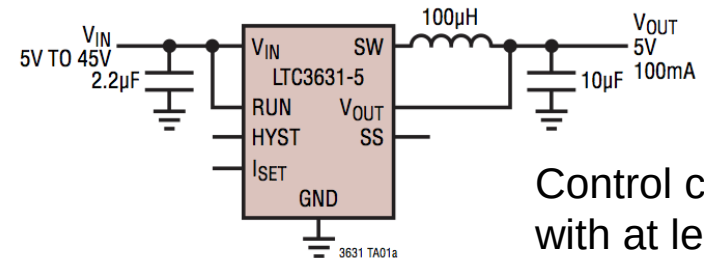
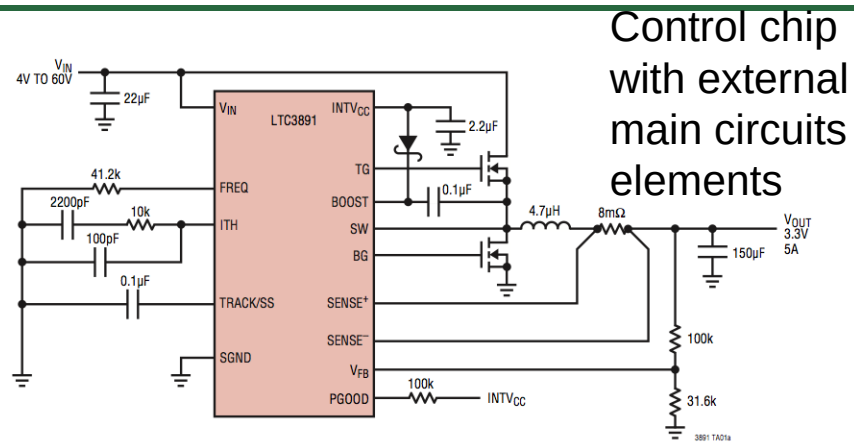
- High efficiency
- Small - no heat sinks!
- **Switching noise**
- **More complicated**

How much more complicated ?



Levels of integration

Controller
Monolithic
 μ Module



Complexity of design

Controller

- Output and input filter
 - Capacitor size and ratings
- Inductor dimensioning
 - Saturation current
 - Core losses
 - Copper losses
- Switches dimensioning
 - Voltage rating
 - Current rating
 - $R_{ds(on)}$ vs Q_g
- Layout
 - **Complex !**

Monolithic

- Output and input filter
 - Capacitor size and ratings
- Inductor dimensioning
 - Saturation current
 - Core losses
 - DC losses
- Layout
 - Medium complexity !

μModule

- Pick the part
 - Fits spec yes/no?
- Layout
 - Piece of cake !



Switch mode supply

What frequency
should I choose ?

48kHz

150kHz

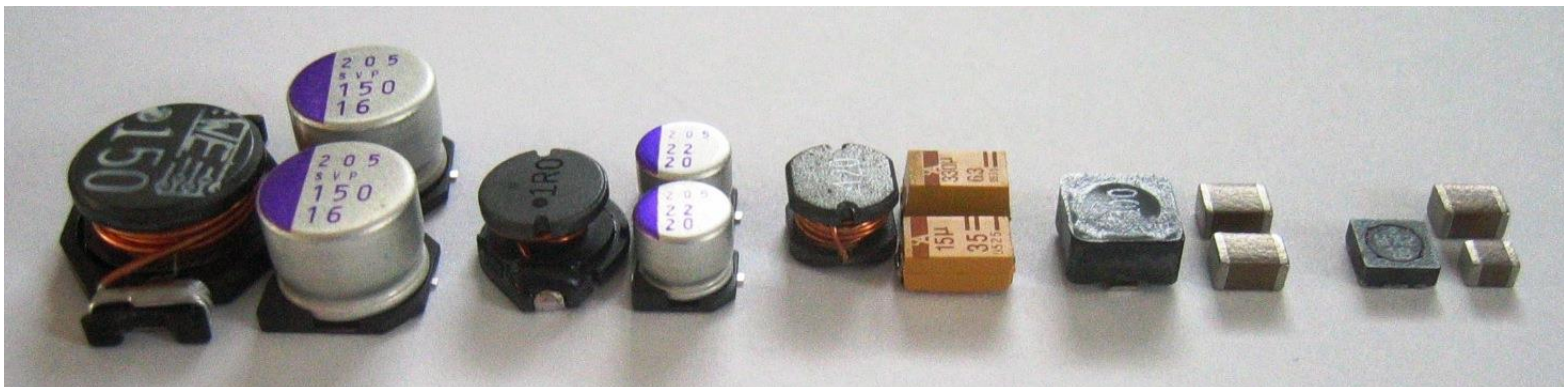
300kHz

500kHz

2MHz

Switching frequency

Switching losses

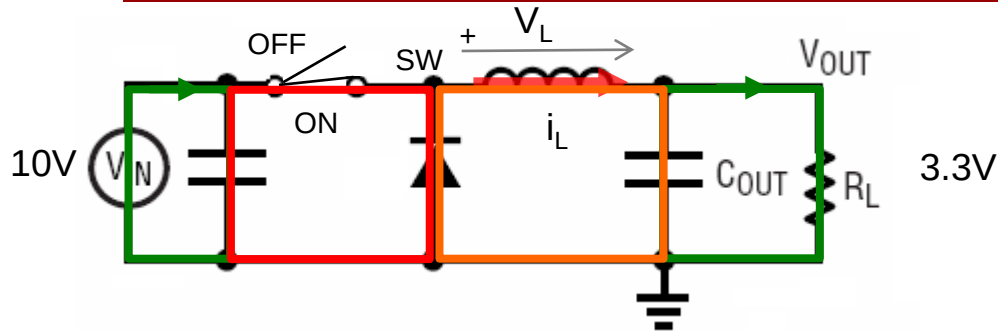


But first some basics



- What is a switch mode regulator?
- Lets have a look at a step down converter (Buck)

Buck Converter Step down

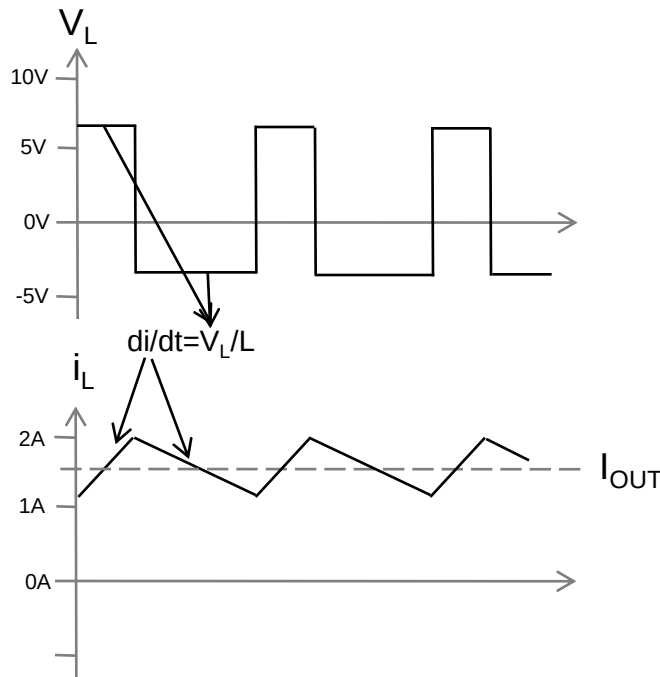
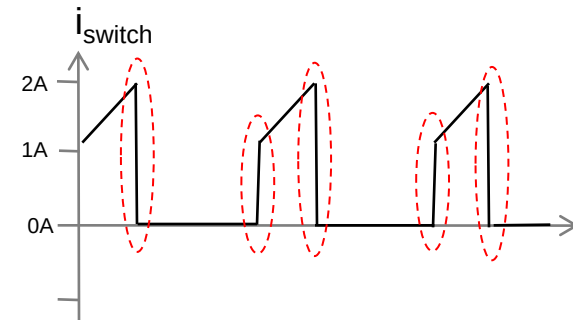
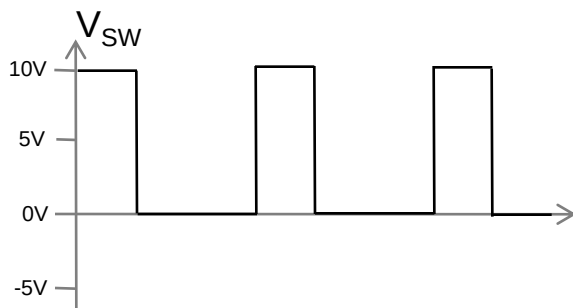


From EMI point of view:

DC currents, no worries...?

AC loop moderate di/dt

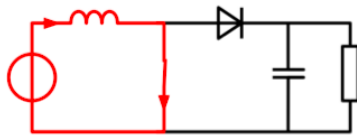
AC loop high di/dt CAUTION !



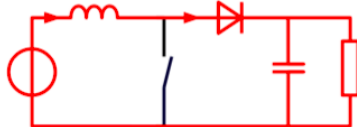
Other topologies

Boost converter

Switch on

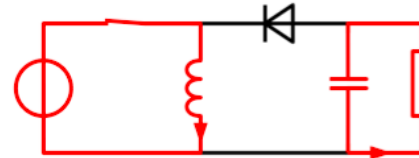


Switch off

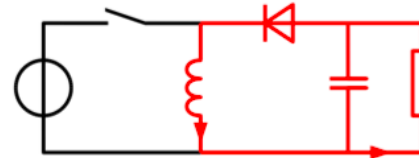


Inverting converter

Switch on

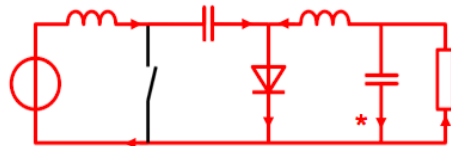


Switch off

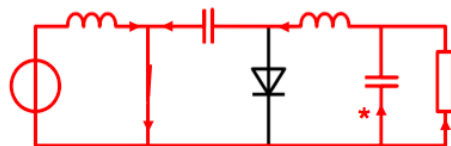


Cuk inverting converter

Switch off

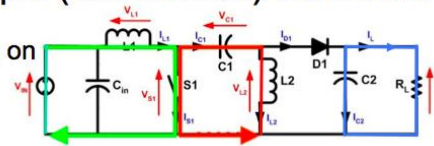


Switch on

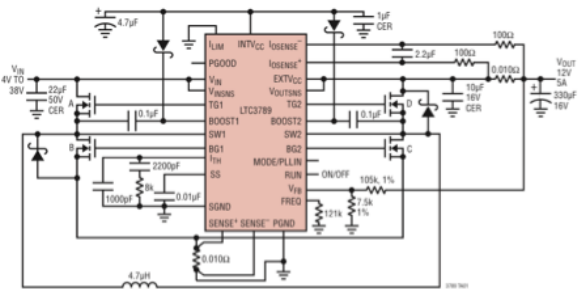
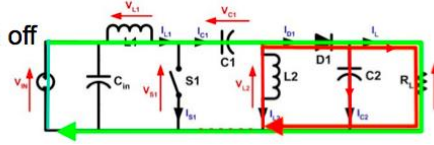


Sepic (buck-boost) converter

Switch on



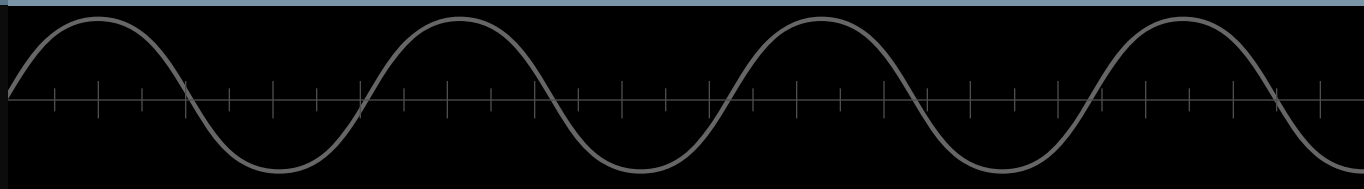
Switch off



+ Isolated converter topologies

EMI and filtering for DC/DC converters

Thomas Ginell
Field Application Engineer
Linear Technology Corporation

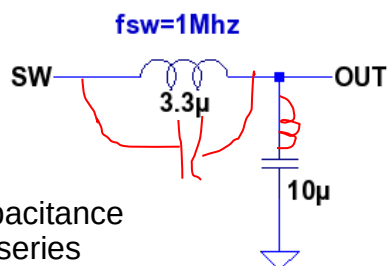
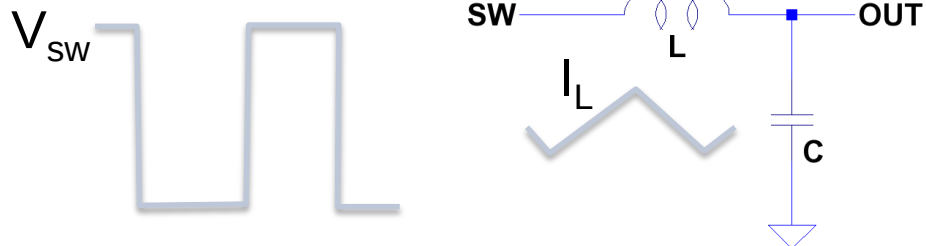


EMI

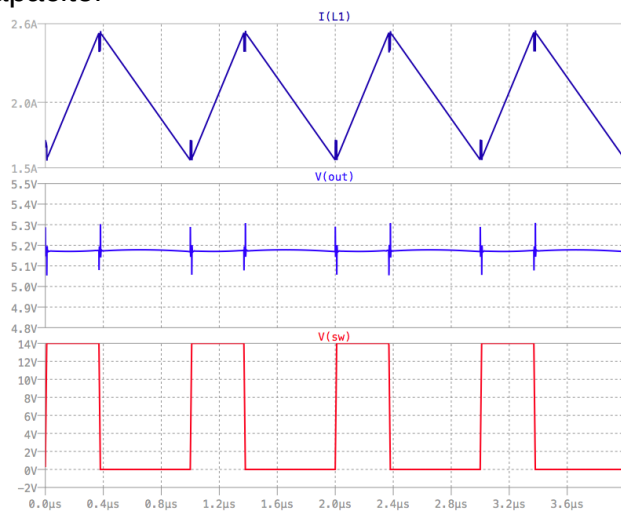
- ◆ Application voltage quality requirements
- ◆ On board interference
- ◆ Emmision
 - ◆ Radiated
 - ◆ Conducted

Output ripple

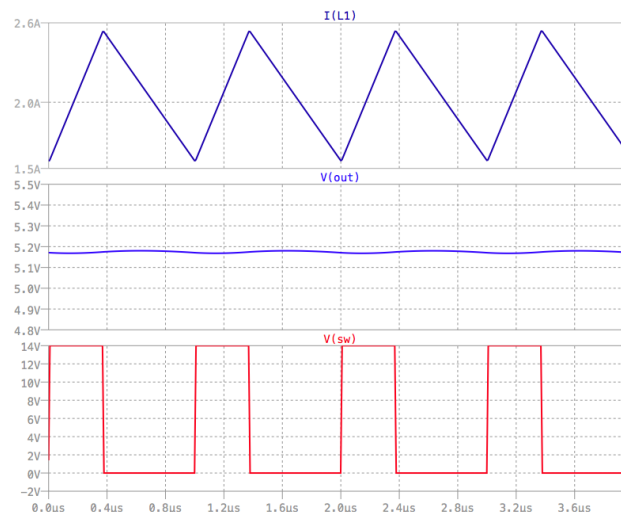
Buck example



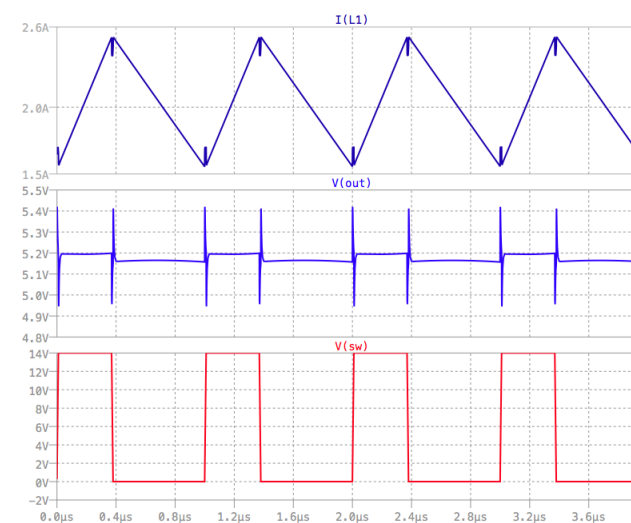
Add parallel capacitance to inductor and series inductance to capacitor



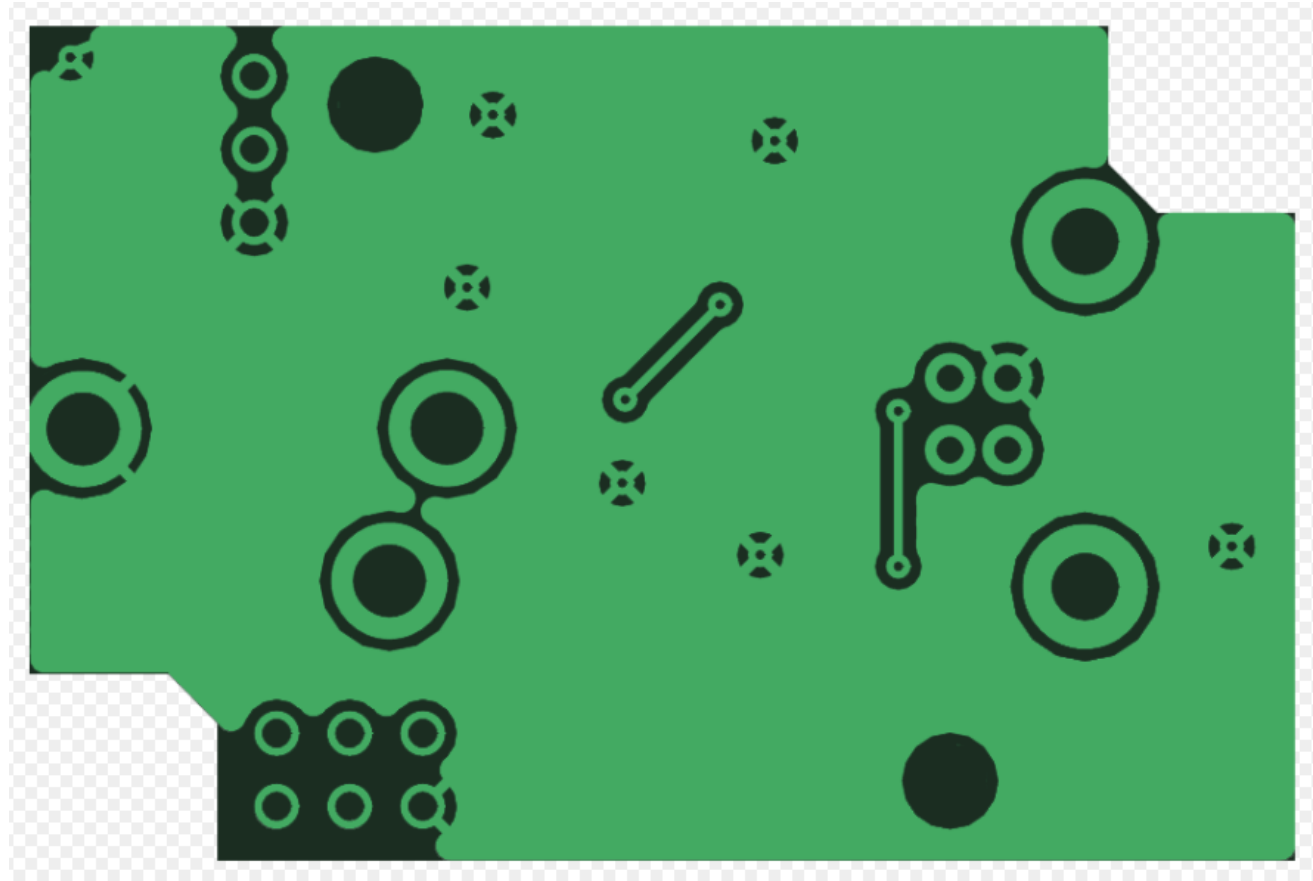
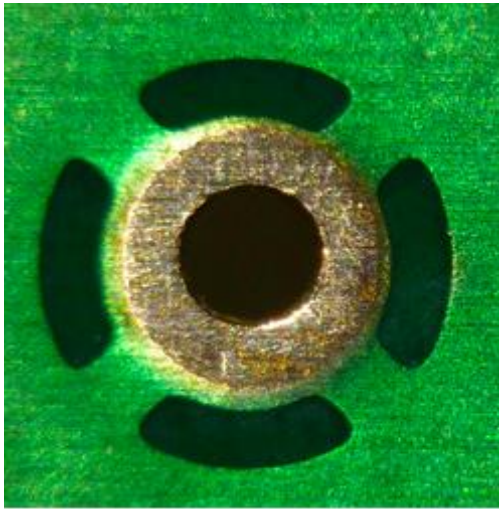
No parasitics waveform (esr 2mohm)



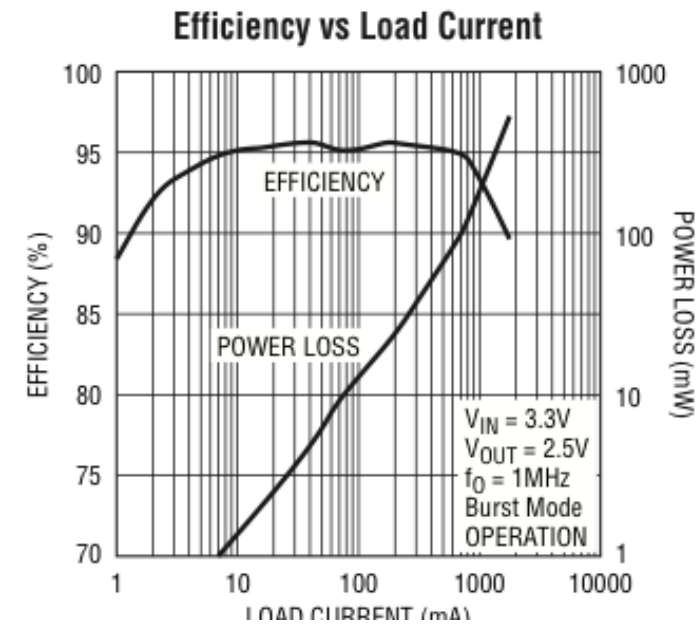
Also use vias and thermal relief for decoupling capacitor



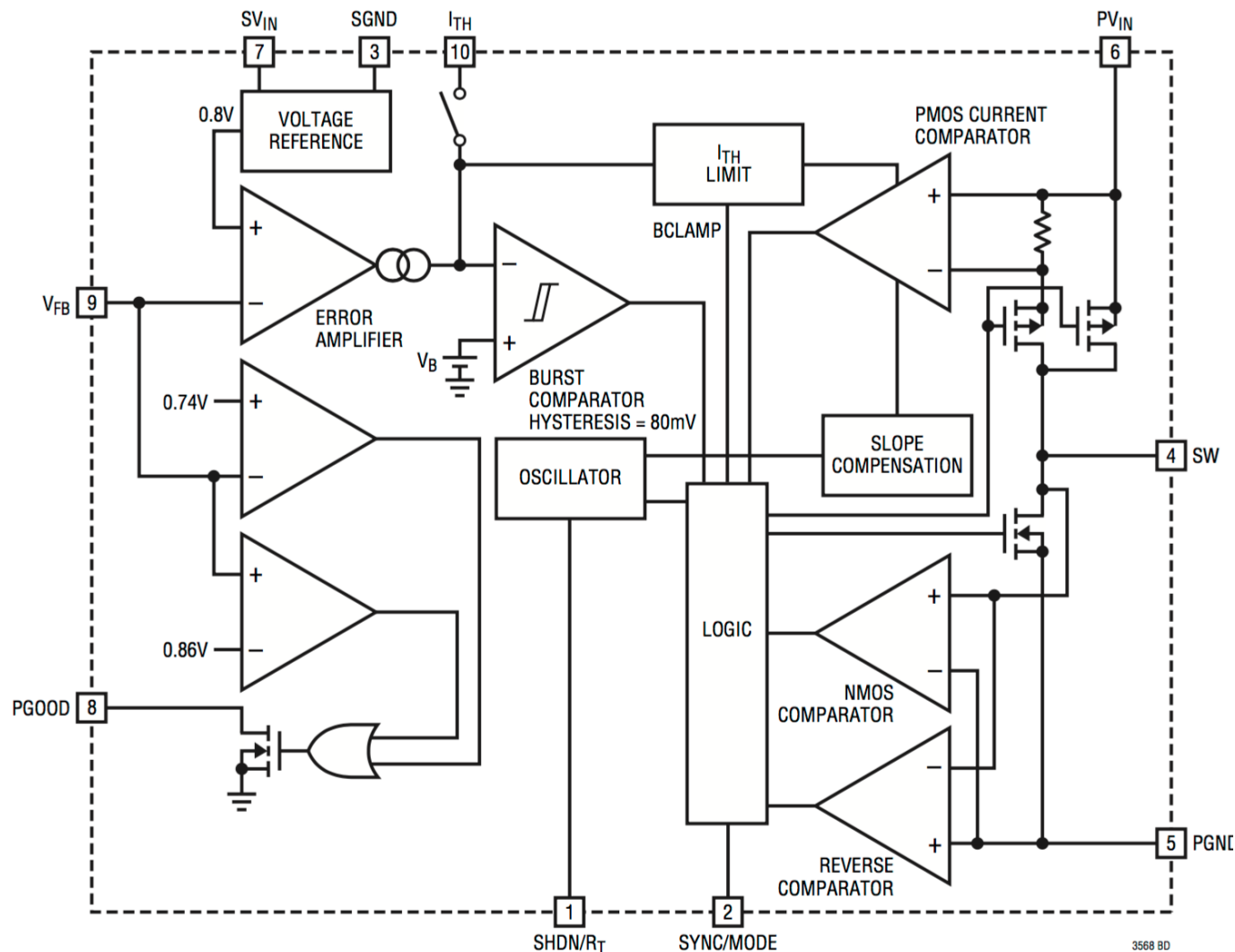
Thermal relief



- ## TYPICAL APPLICATION



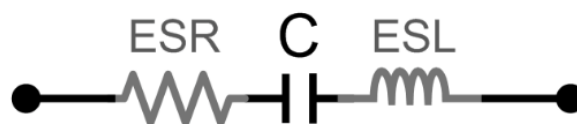
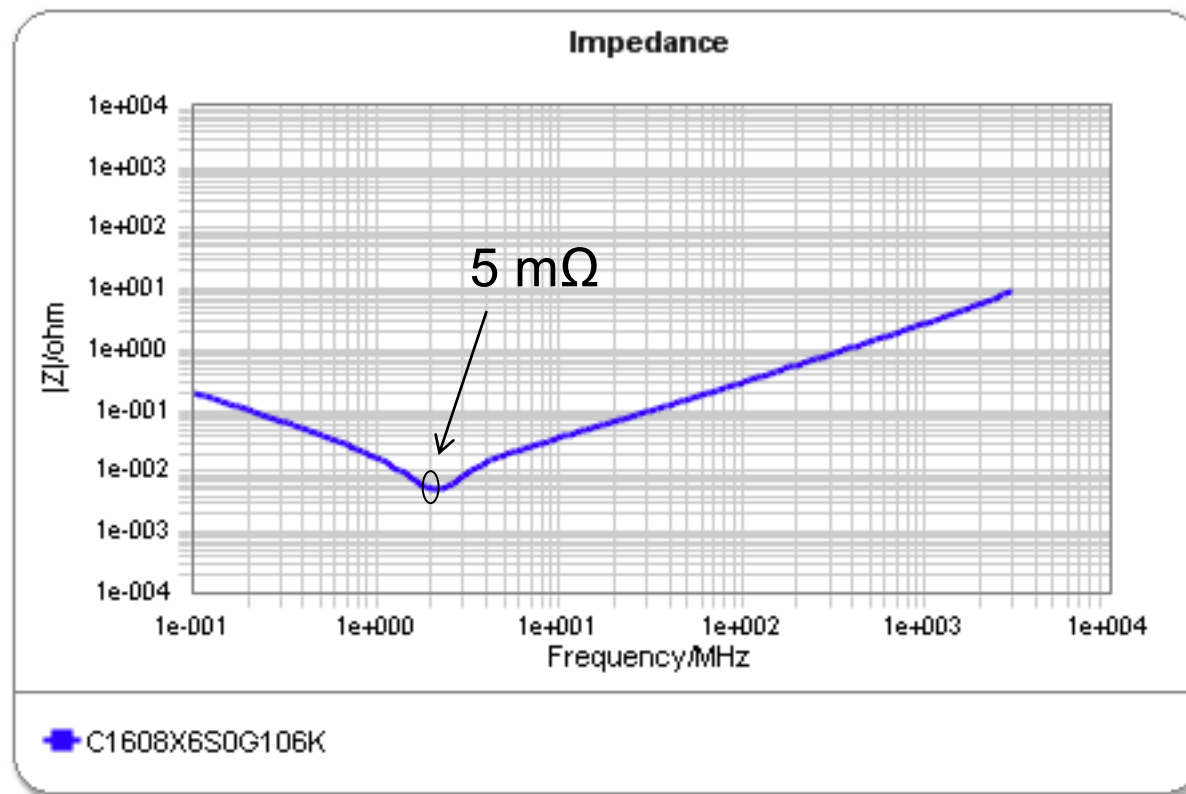
LTC3568 block diagram



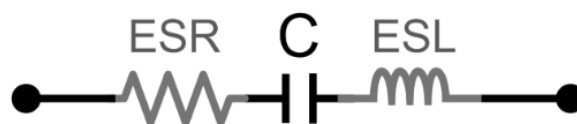
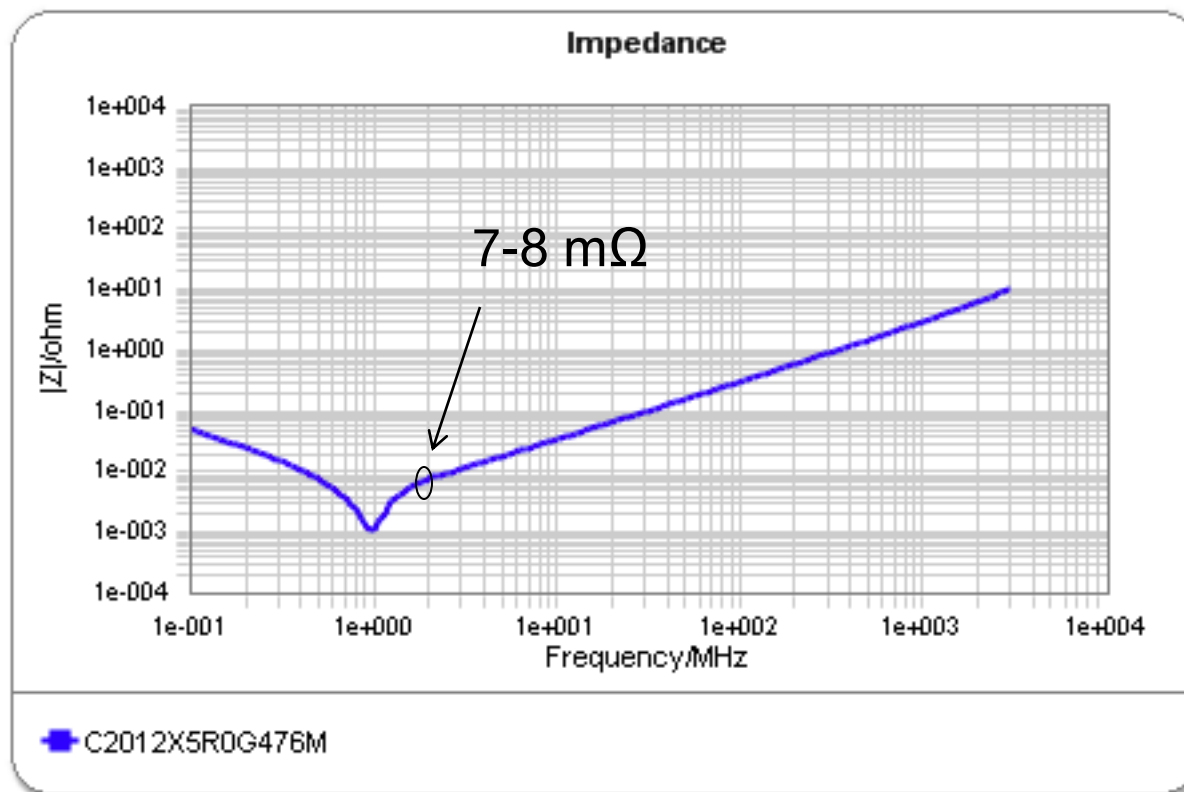
Chose output capacitor

	<i>comparison</i>	Metric code	Imperial code	output
• 0201 (0603 metric)	0.1x0.1 mm	0402	01005	
• 0402 (1005 metric)		0603	0201	
• 0603 (1608 metric)		1005	0402	!
• 0805 (2012 metric)		1608	0603	
• 1008 (2520 metric)	1x1mm	2012	0805	
• 1206 (3216 metric)		2520	1008	
• 1210 (3225 metric)		3216	1206	!
• 1806 (4516 metric)		3225	1210	
• 1812 (4532 metric)		4516	1806	
• 2010 (5025 metric)		4532	1812	
• 2512 (6432 metric)	1x1 cm	5025	2010	
		6332	2512	
		Actual size		

4V 10 μ F X6S

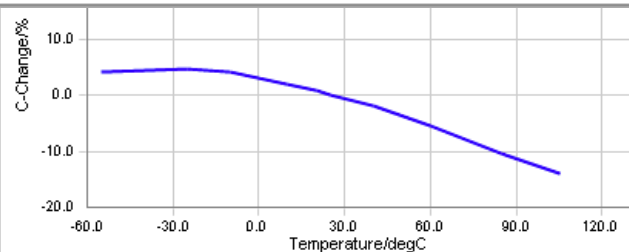
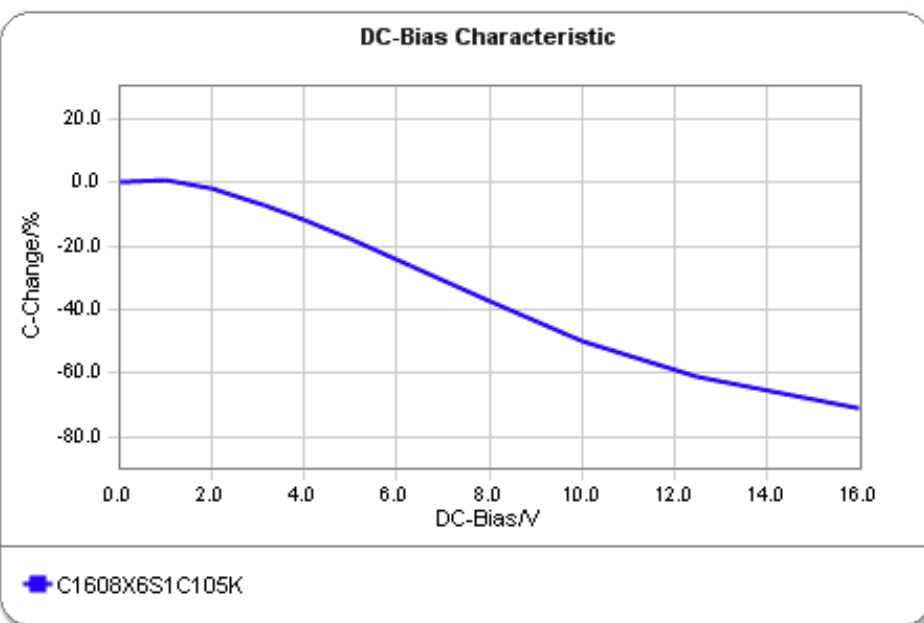


4V 47uF X5R

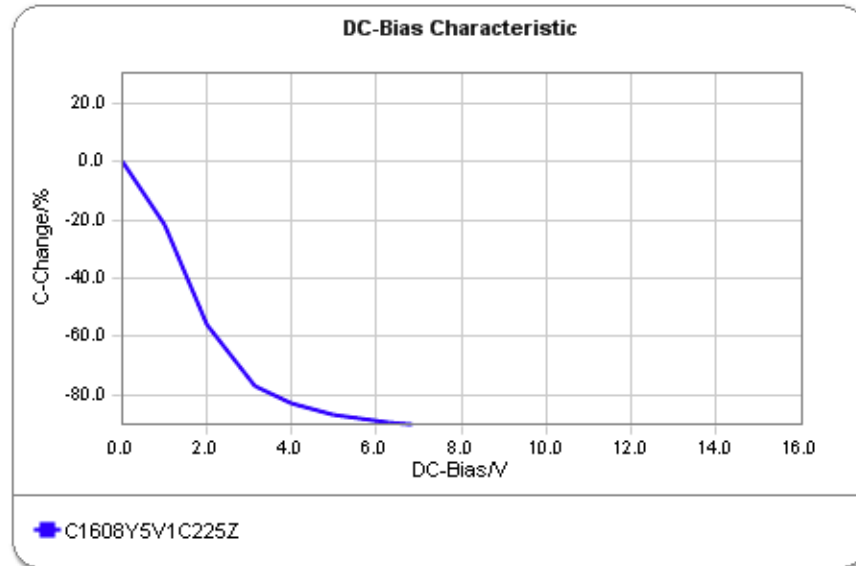


Beware ! Bias dependency !

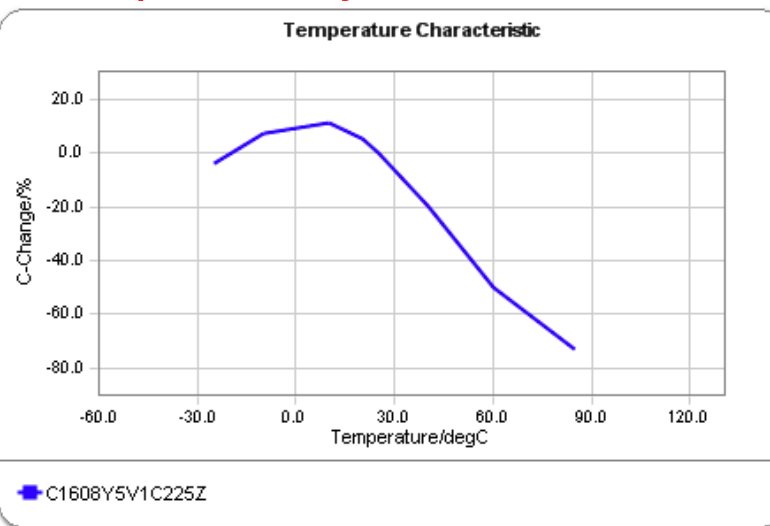
16V X6S +/- 10%



16V Y5V +80% -20%

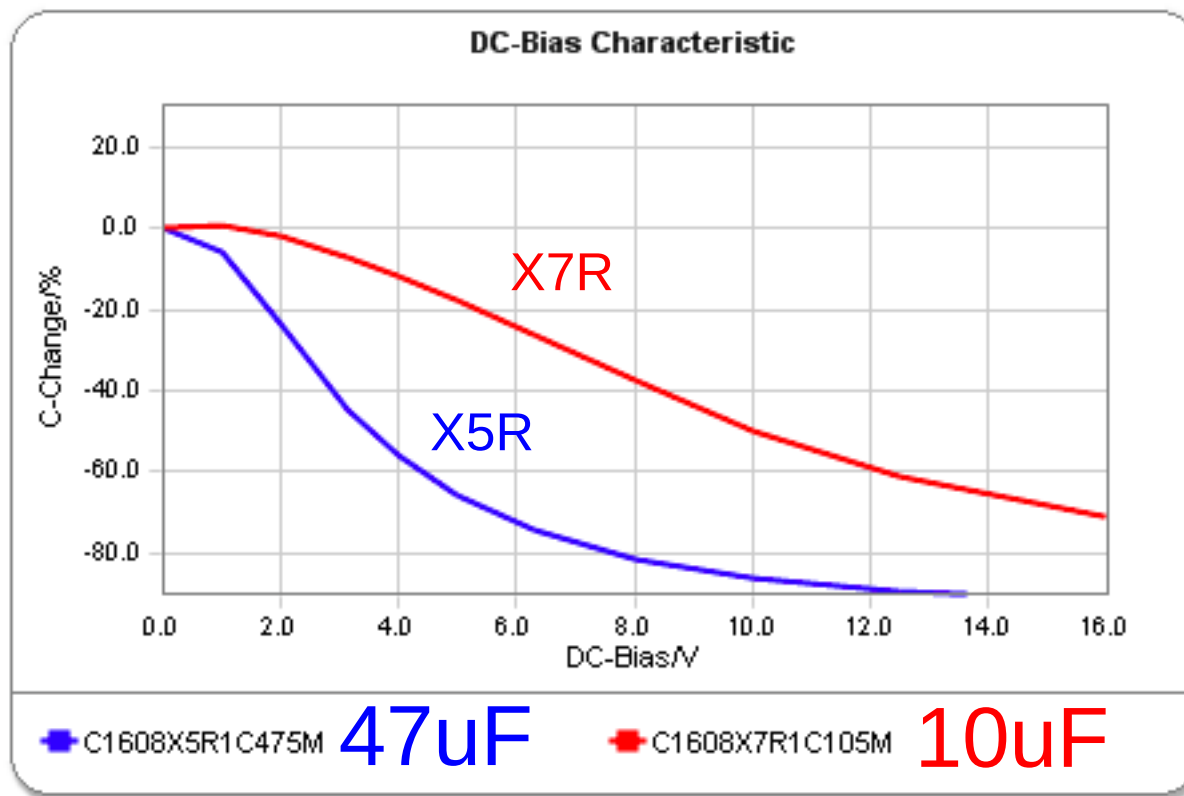


Then add temperature dependency



X7R vs X5R

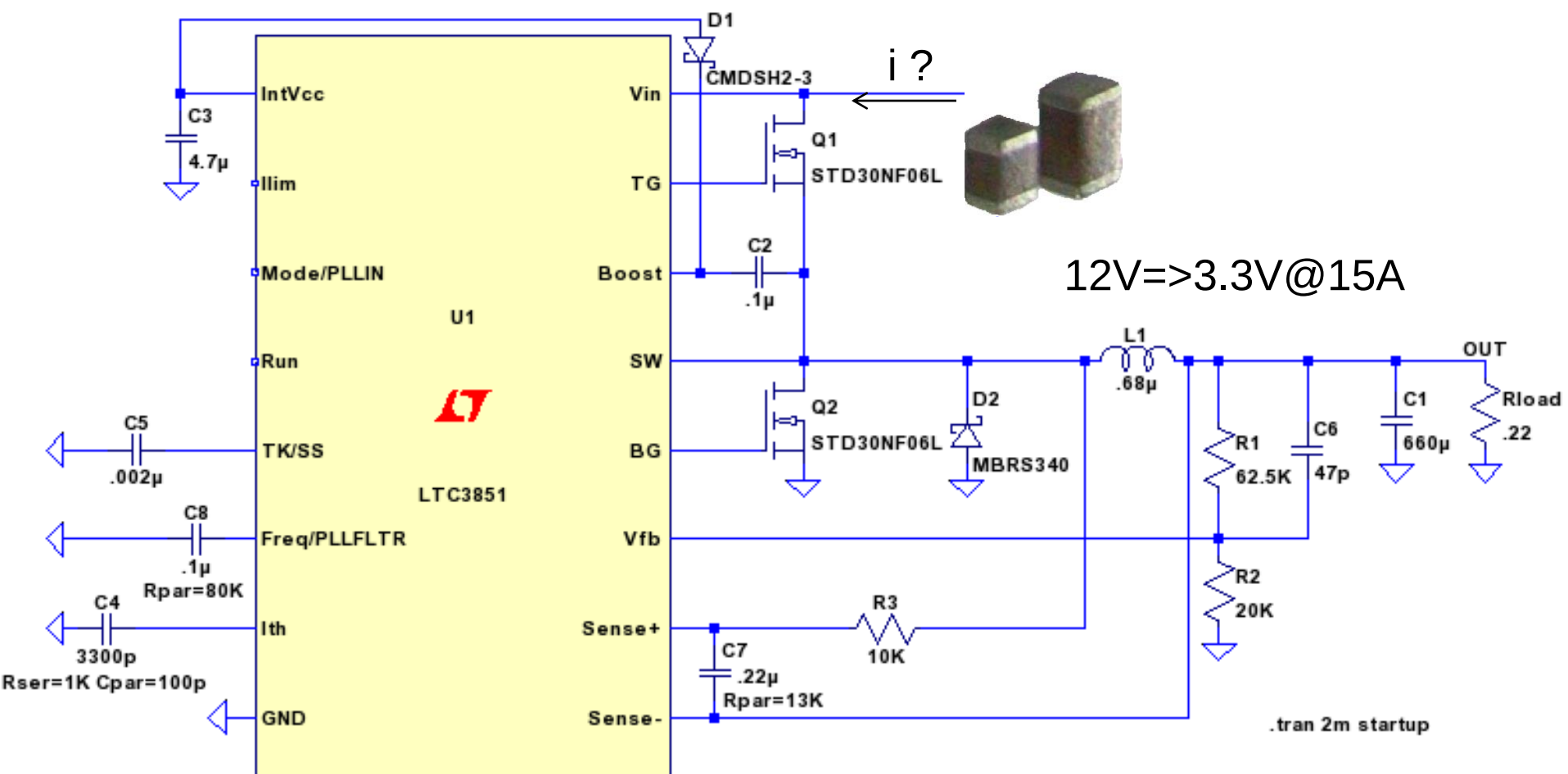
Same size, 0603



BUT, the X5R cap is almost 5 times larger to start with
X5R cap still almost 2x capacitance at 5V !

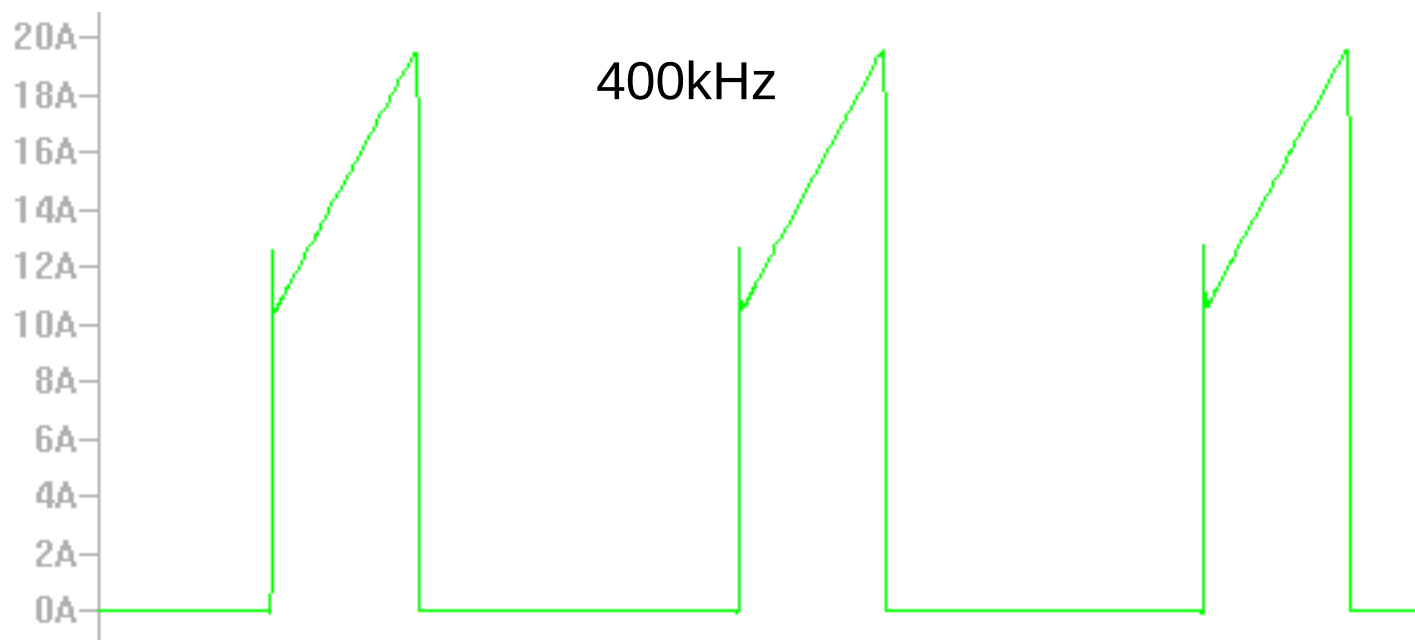
Input ripple

Buck example



Buck input current (from decoupling cap)

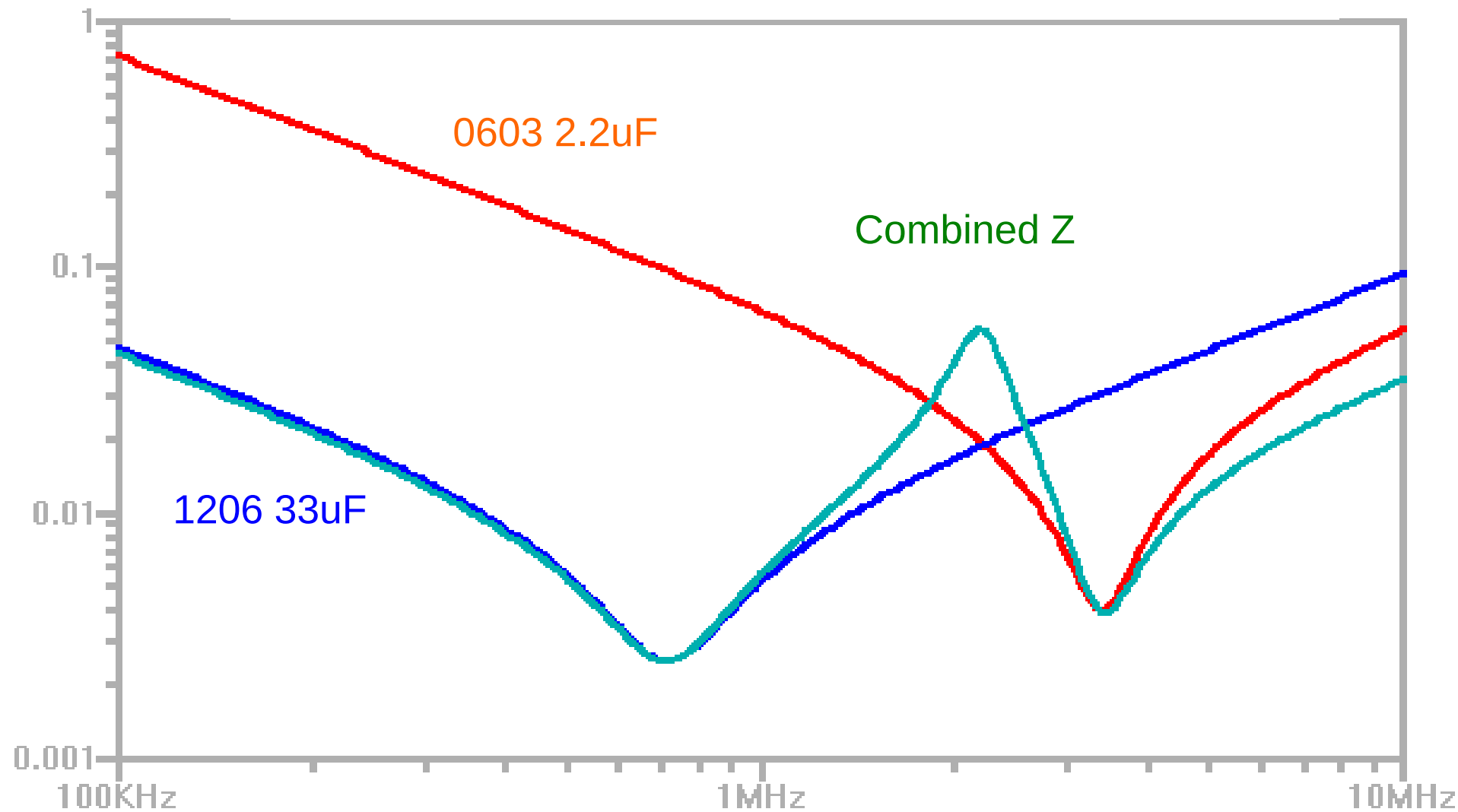
12V=>3.3V@15A



Decoupling

- ◆ Small ceramic capacitor:
 - ◆ 0603 25V 2.2uF esr 4mΩ L=1nH
- ◆ Large ceramic capacitor
 - ◆ 1206 25V 33uF esr 2.5mΩ L=1.5nH

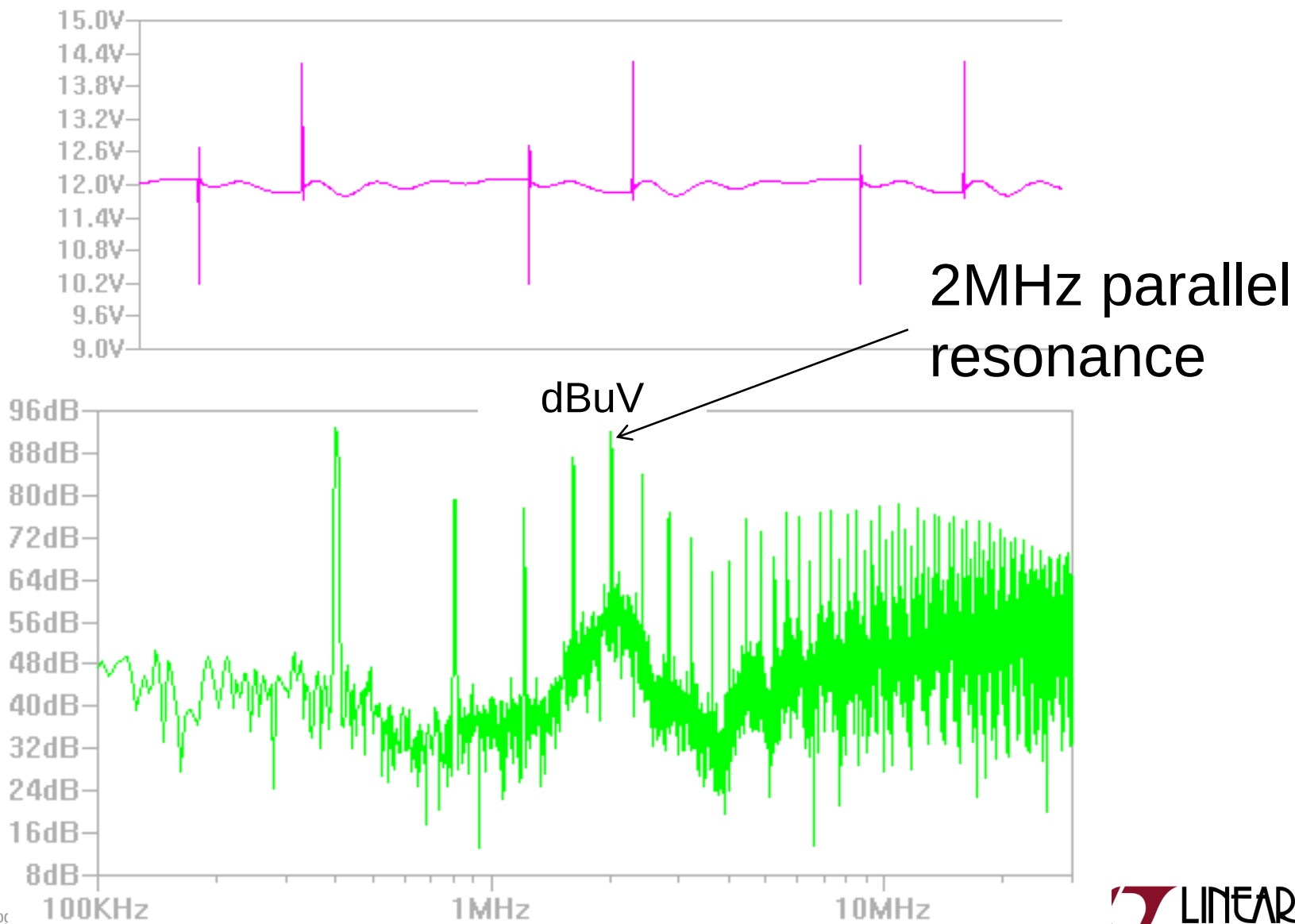
Filter capacitor Z



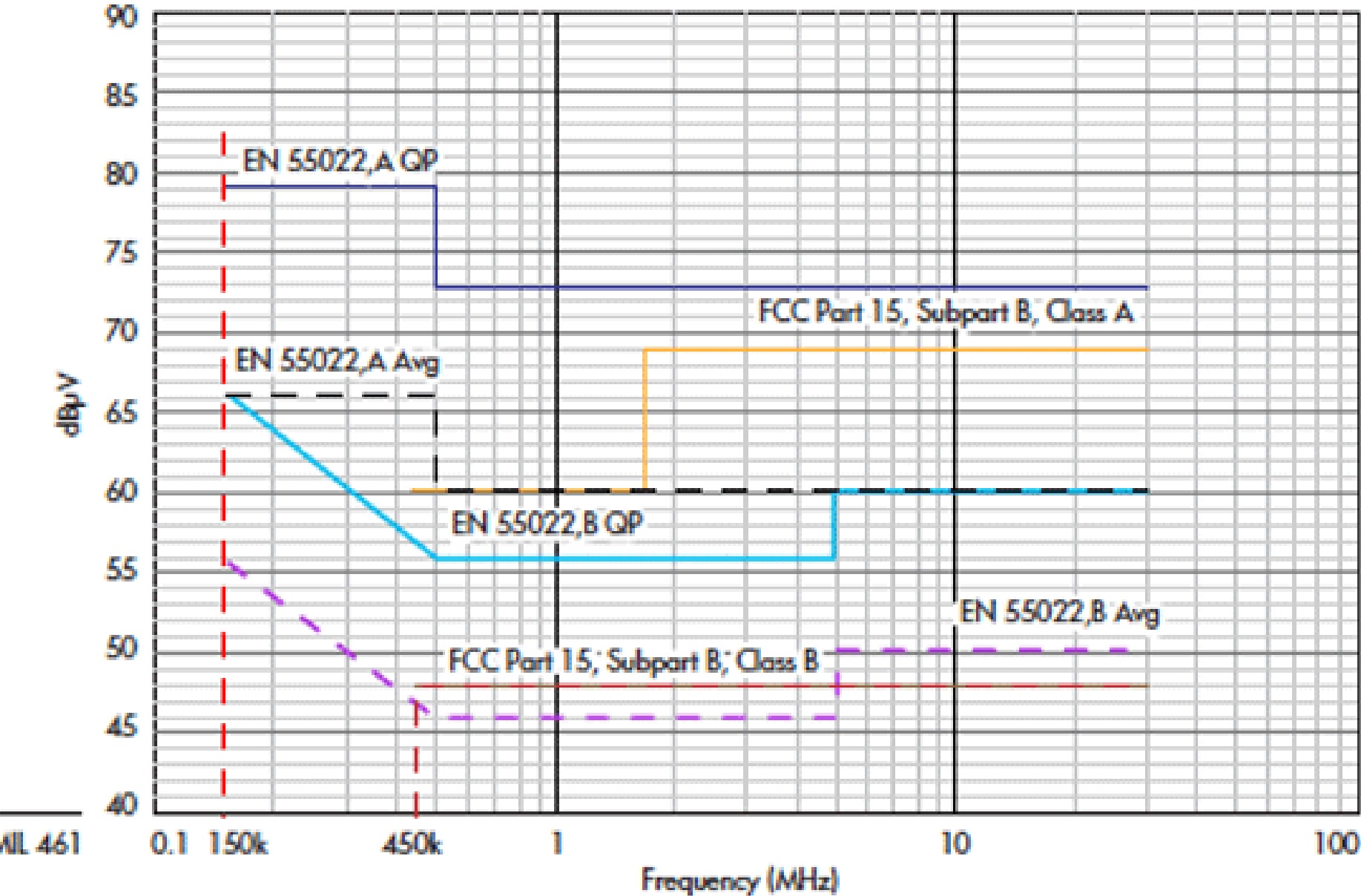
Filter capacitors Z

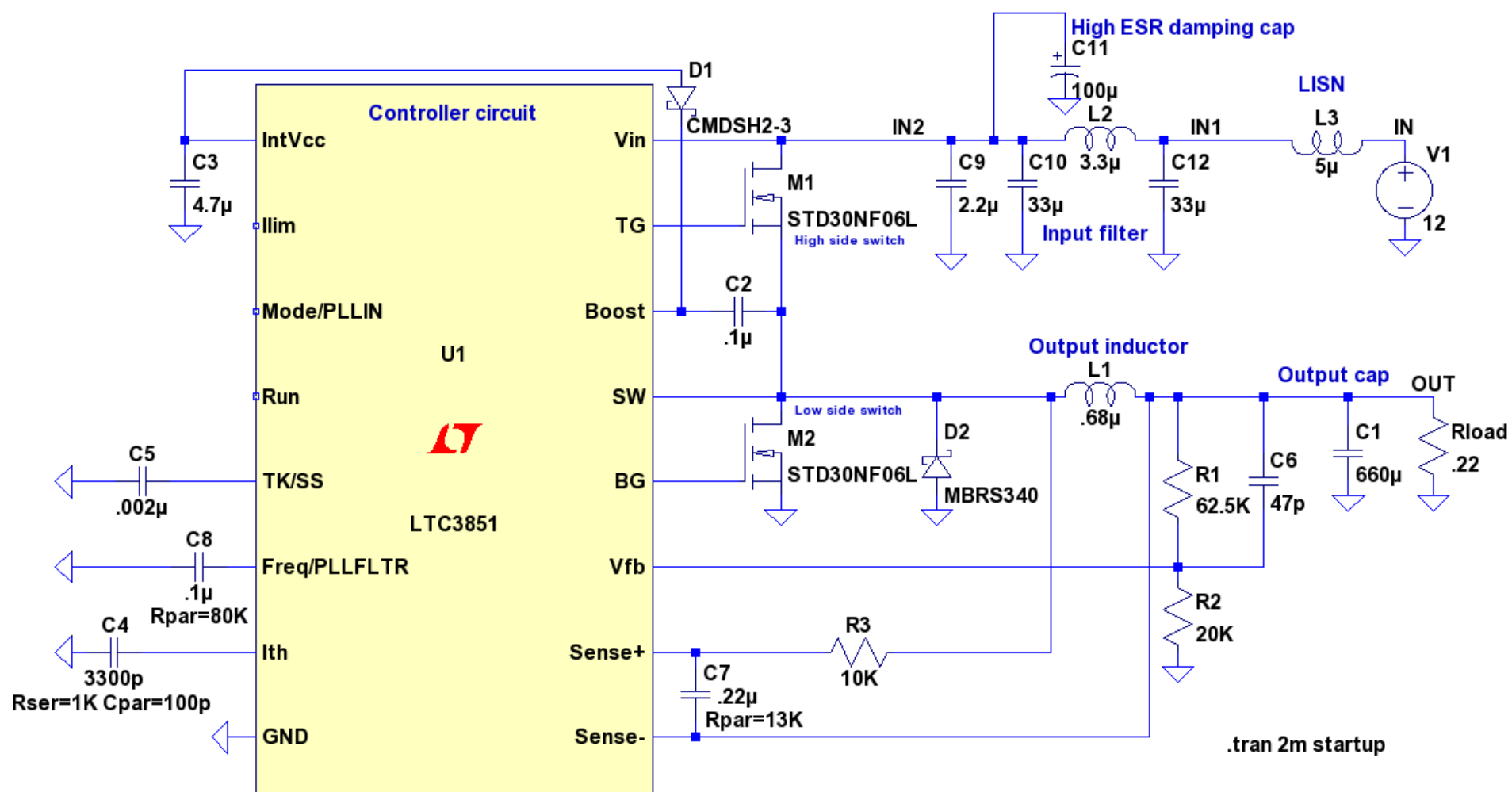
- ◆ Use multiple package and capacitor sizes to combine their Z:s
 - ◆ Above might not be true, be smart?!?
- ◆ Its easy to simulate the low frequency behavior for EMI purposes
 - ◆ REALLY !
 - ◆ (150kHz-30MHz for EMI, higher for Decoupling)
 - ◆ LTspice from Linear Technology
 - ◆ Free
 - ◆ Easy to use
 - ◆ The most used spice simulator in the world
 - ◆ Fast, stable, accurate

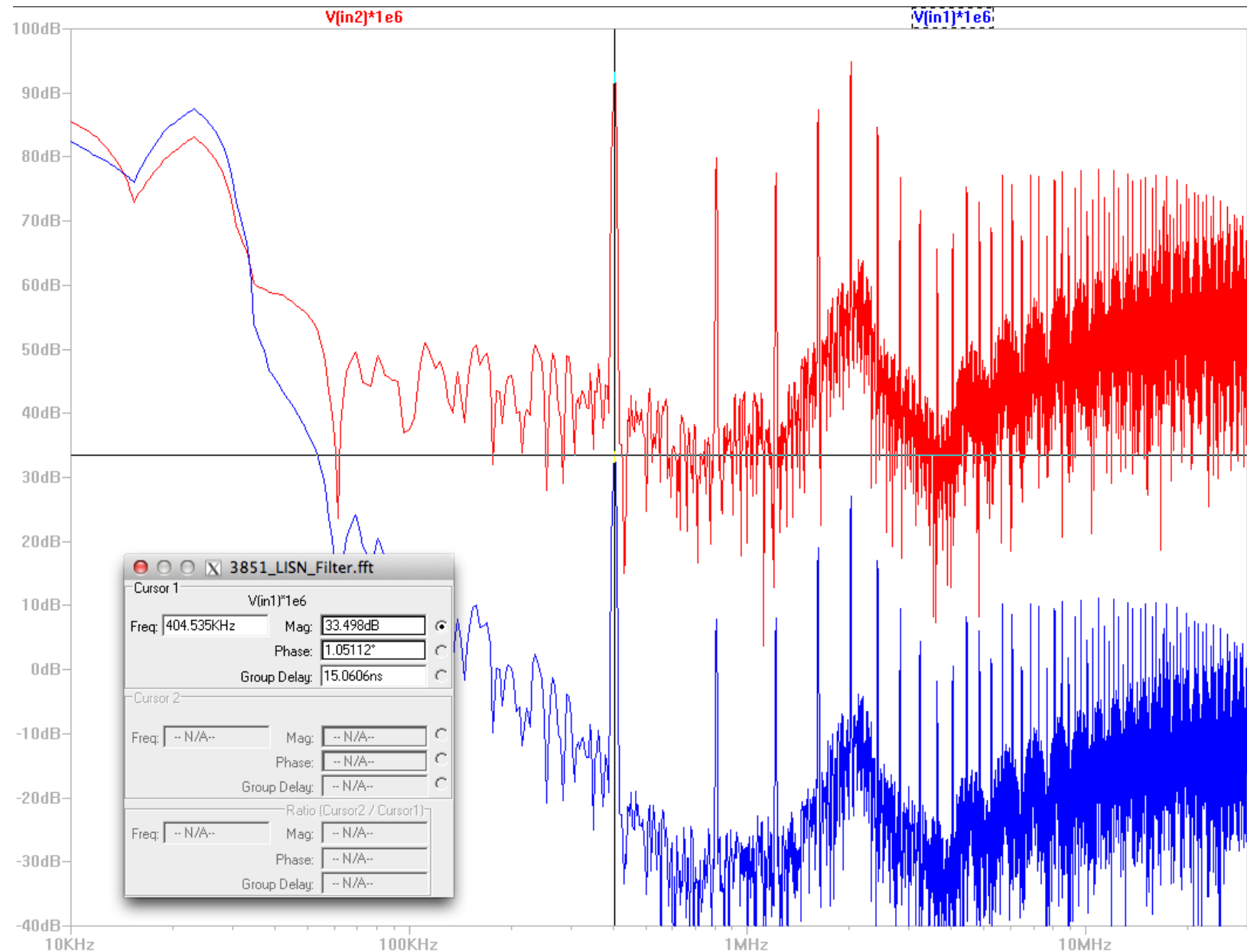
How does the input voltage look on the input cap



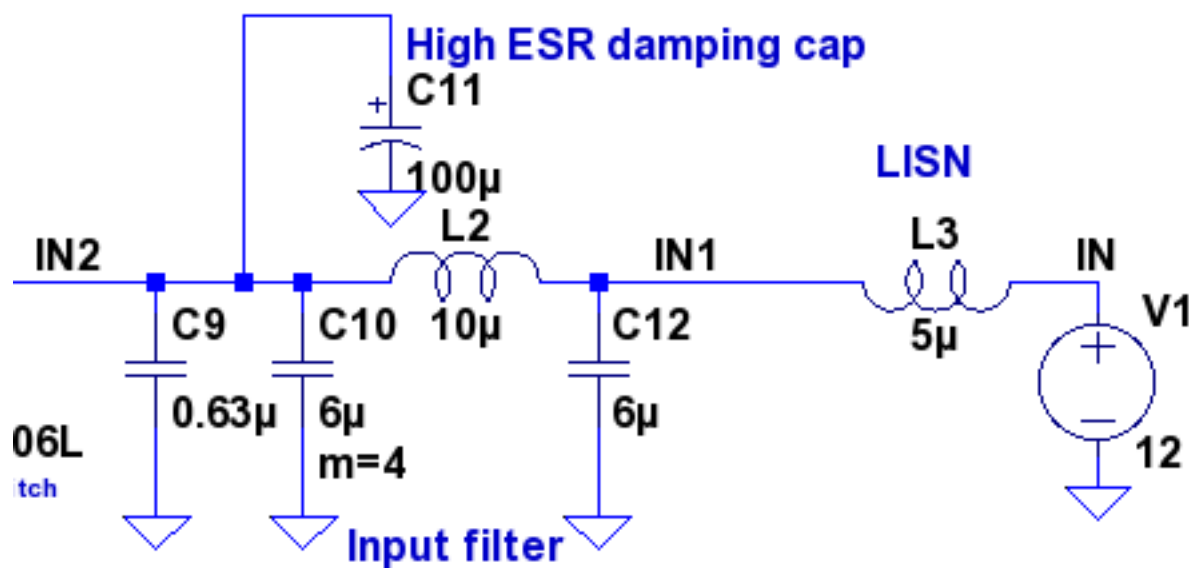
Standard limits



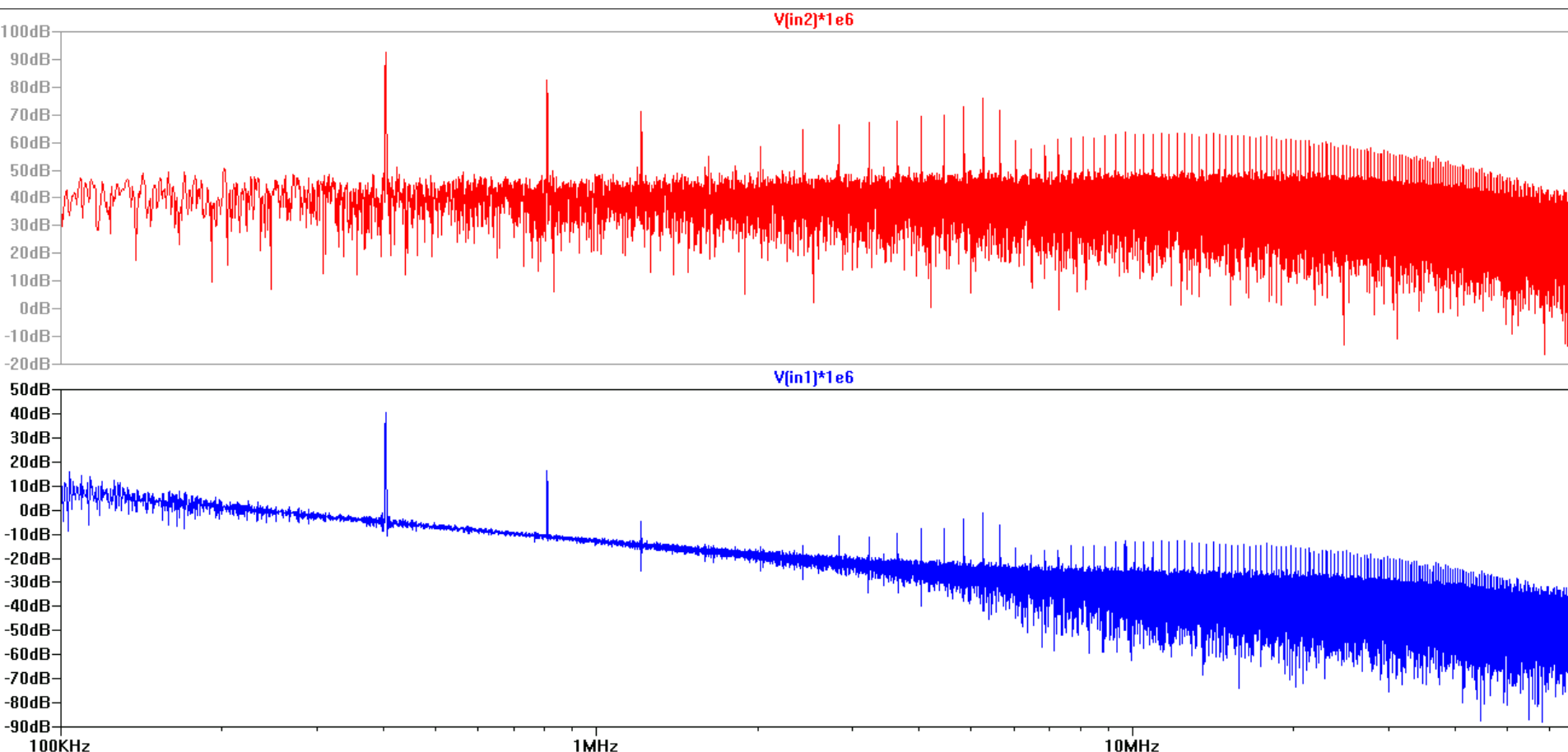




But you need to use DC bias derating

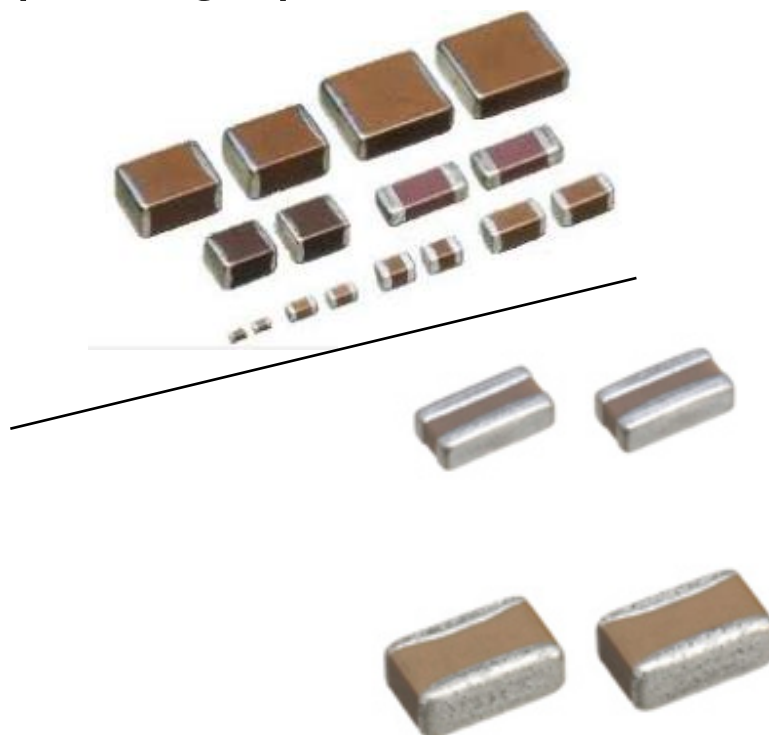


New result



Decoupling strategy

- ◆ Keep impedance low over frequency
 - ◆ Beware of parallel resonance !
- ◆ Some capacitor package parasitic inductances
 - ◆ 1210 0.80nH
 - ◆ 0806 0.70nH
 - ◆ 0603 0.60nH
 - ◆ 0402 0.42nH
 - ◆ 0612 0.20nH
 - ◆ 0306 0.10nH
 - ◆ 0204 0.07nH
 - ◆ 3025 1.30nH



Layout in Deep

Switching Power Supply PCB Layout

An In Deep Look

Edited and Presented by Thomas Ginell
FAE Scandinavia

Relevant Physical Effects in Switchmode PSU

1. Skin effect
2. Proximity effect
3. Magnetic dipol antennas
4. Near field and far field effects
5. Shielding with planes, Vias
6. Energy distribution over frequency of the different loops in a switch mode PSU
7. Current flow in typical buck and boost
8. Placement and routing strategies
9. Magnetic properties of components

Inductance

Most problems on PSU layouts relate to inductance on places where you do not want it. There are only a very few places where inductance is needed.

This presentation centers how we can support the nature in its inductance avoidance strategies.

Skin effect

As lower resistance material is used as thinner the skin depth gets.

$$\delta = \sqrt{\frac{\rho}{\pi f \mu_r \mu_0}}$$

	Resistivity (Ohm * m)	1Mhz (um)	100Mhz (um)
Copper	1,68e-8	66	6,6
Tin	1,09e-7	166	16,6
10e20 doped silicon	1e-5	1600	160
uMetal	1e-7	<1	<0,1

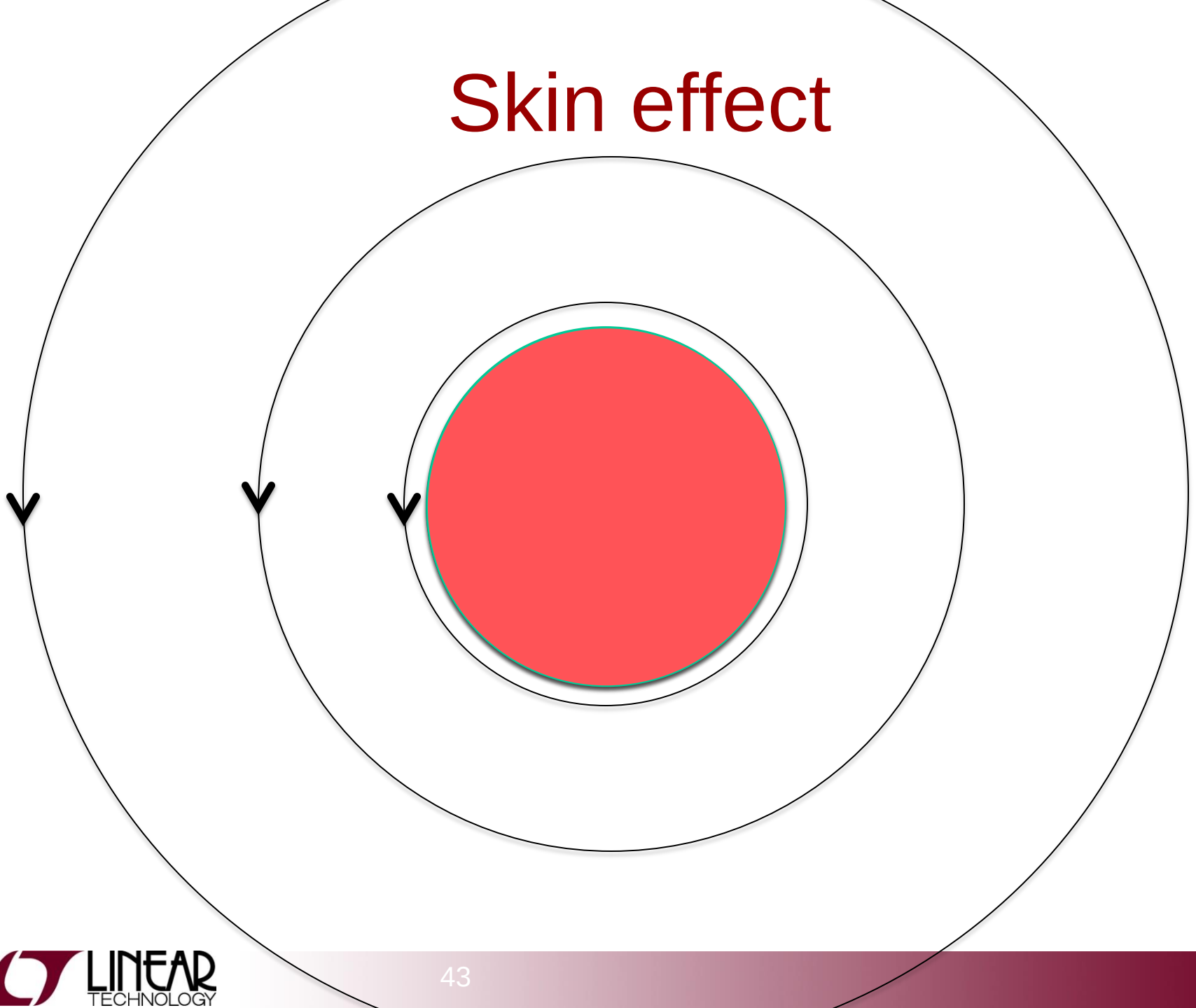
Rule 1.1

AC current will only flow at
the outside of the conductor

Rule 1.4

On 35um (1oz) copper the
skin effect kicks in at about 3,5 Mhz

Skin effect

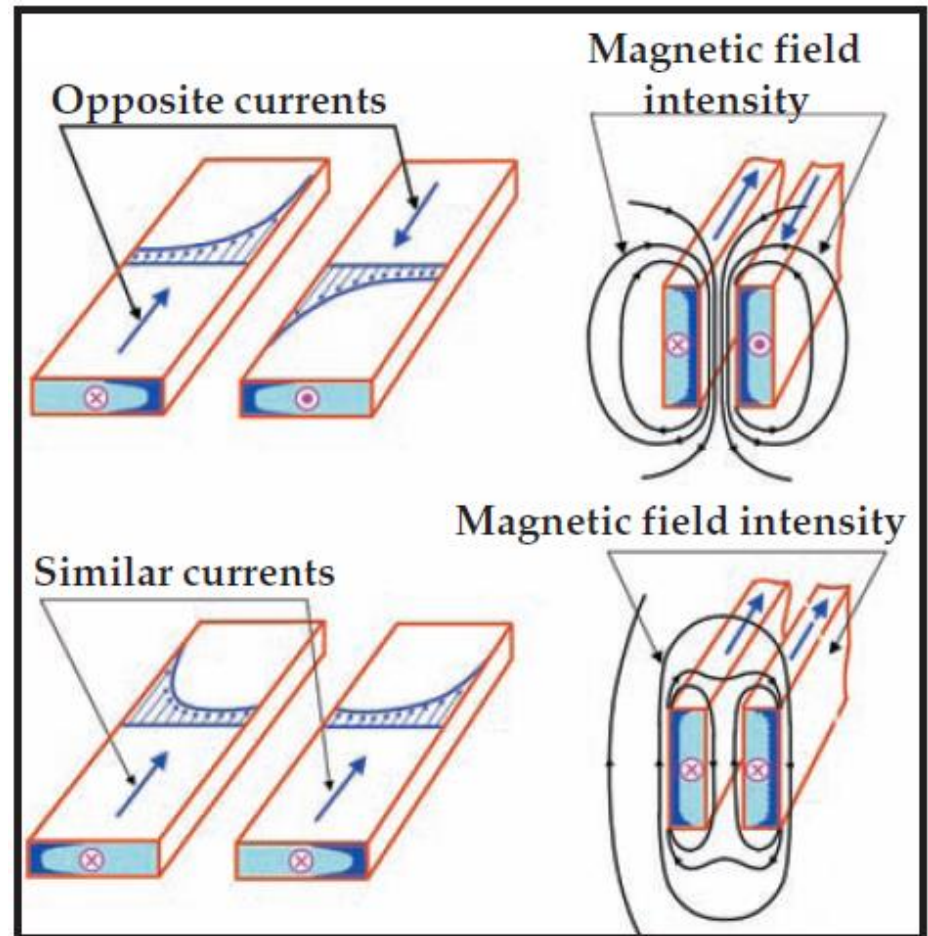


Proximity effect

Currents in the opposite direction attract each other. The same induction forces responsible for the skin effect work here in the opposite way, since the current has opposite direction.

This is the typical case for currents in switchmode PSU layouts.

Currents in the same direction force themselves apart as they do inside the same conductor by the skin effect.



[1]

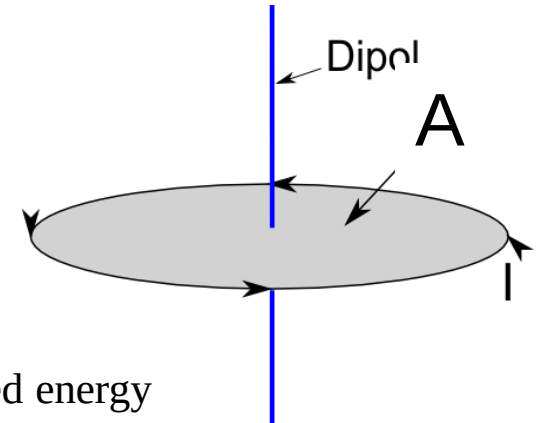
Magnetic dipol antennas

$$R_r = \frac{320 \pi^4 f^4 A^2}{c^4}$$

Since R_r is low against 377 Ohm because loop pattern sizes are low against wavelength in practical switchmode PSUs, the radiated energy will rise proportional with R_r

In Any loop your AC current will circle around, the radiated energy will increase proportional to:

f^4 and A^2



On the frequency f you can not do very much about since any filtering will reduce efficiency.

But on the loop opening F , usually isolated area on your PC Board, you can keep it as small as possible.

Rule 3.1

Emitted radio energy increases with the square of the loop area you create

Rule 3.2

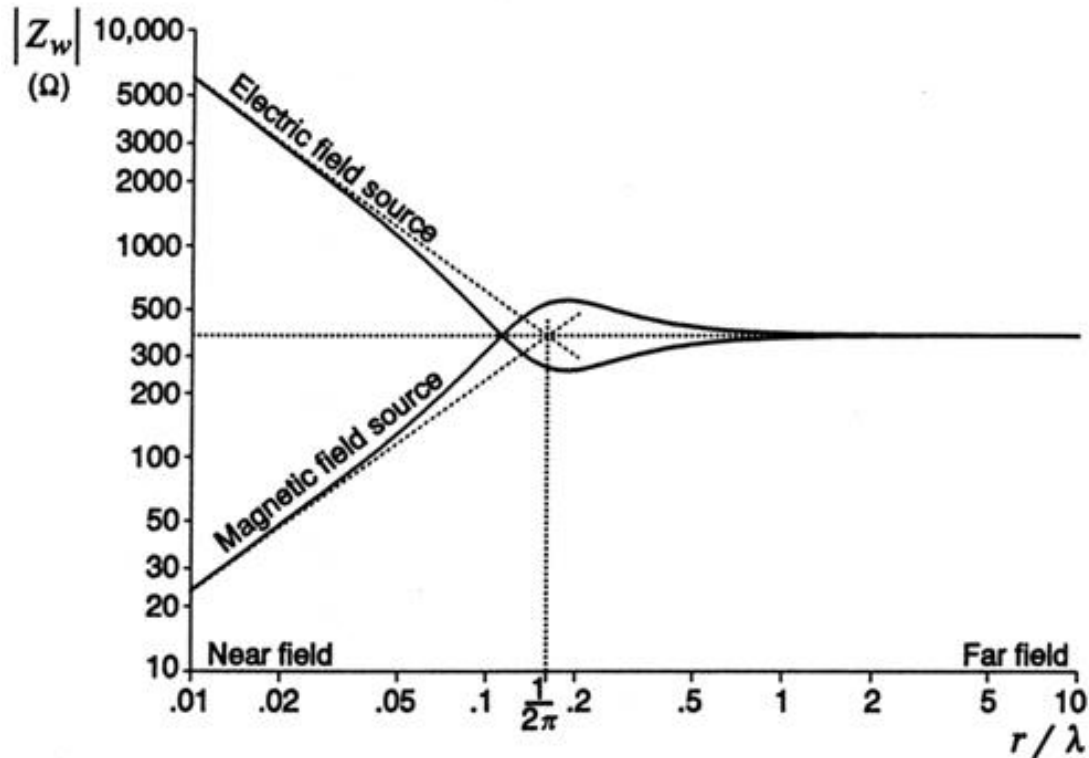
Passive antennas have the same characteristic receiving and transmitting.

Near field versus far field

Any near field transforms itself to a far field pattern

at the distance of about $\frac{\lambda}{2\pi}$ and adapts the 377 Ohm

far field impedance of our Universe.

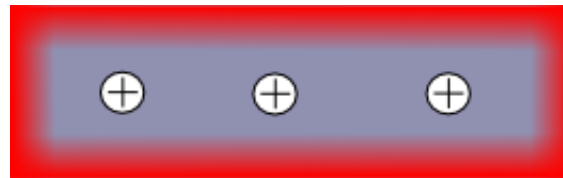


[5]

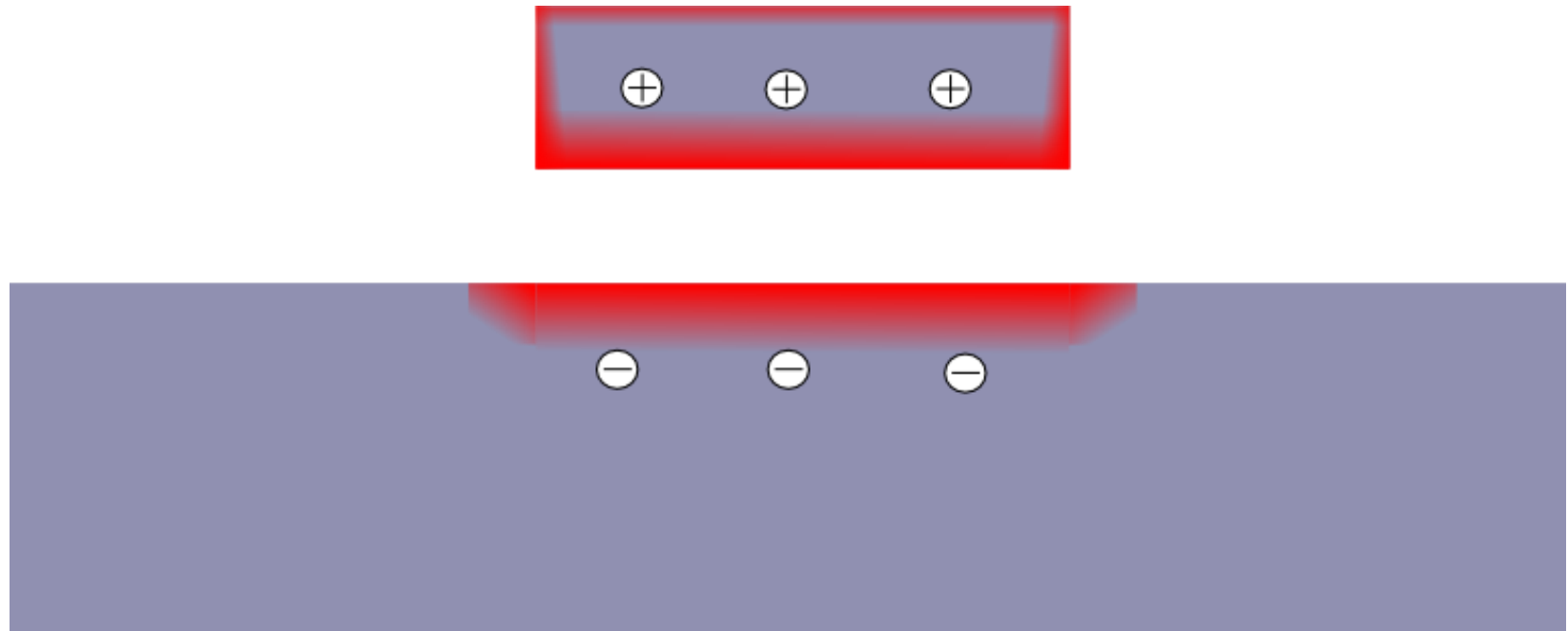
Rule 4.1

Magnetic AC fields will always find a way to turn into radio waves out of your box.
So keep them as small as possible

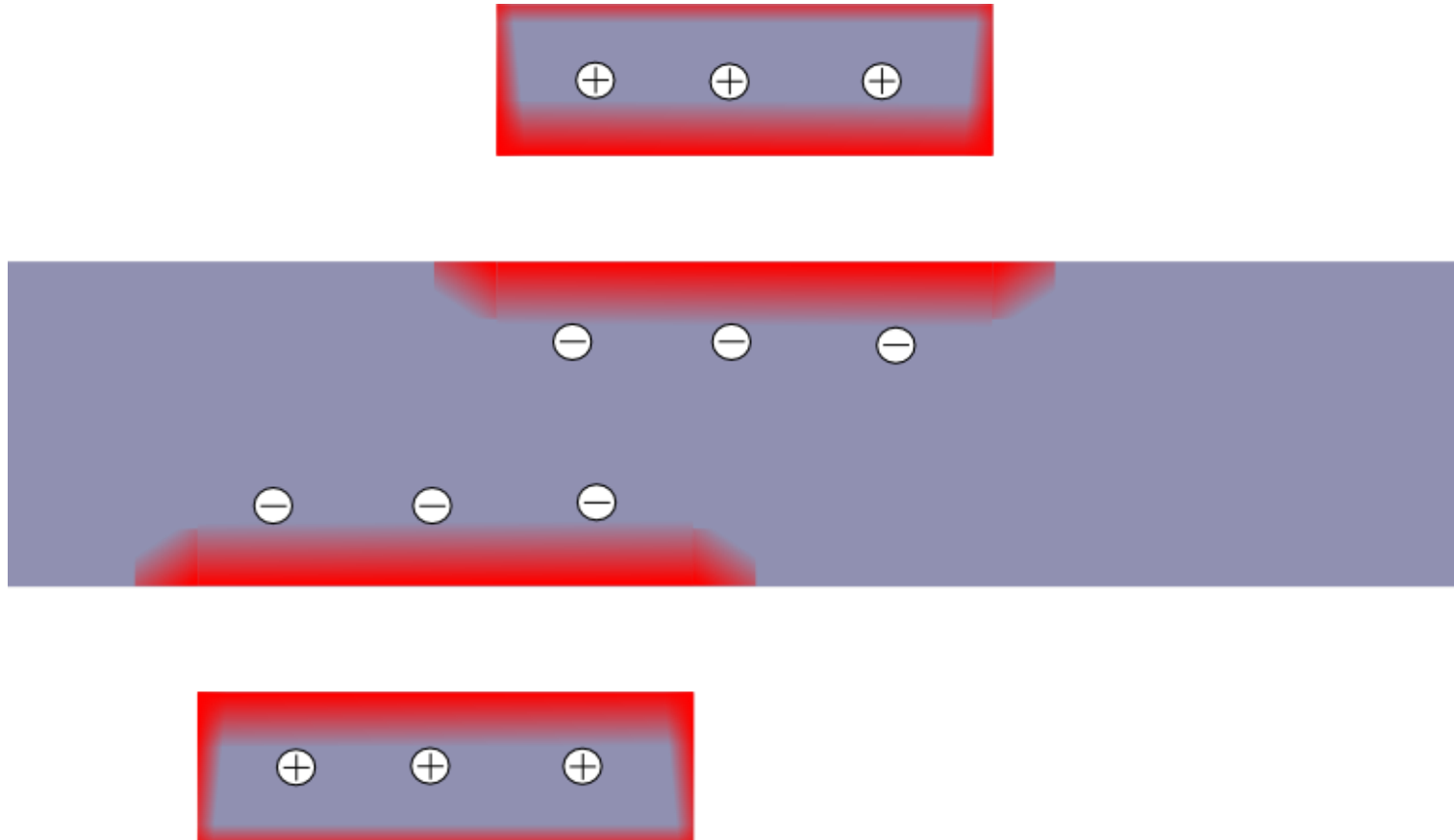
Rectangular conductor alone



Return current in plane

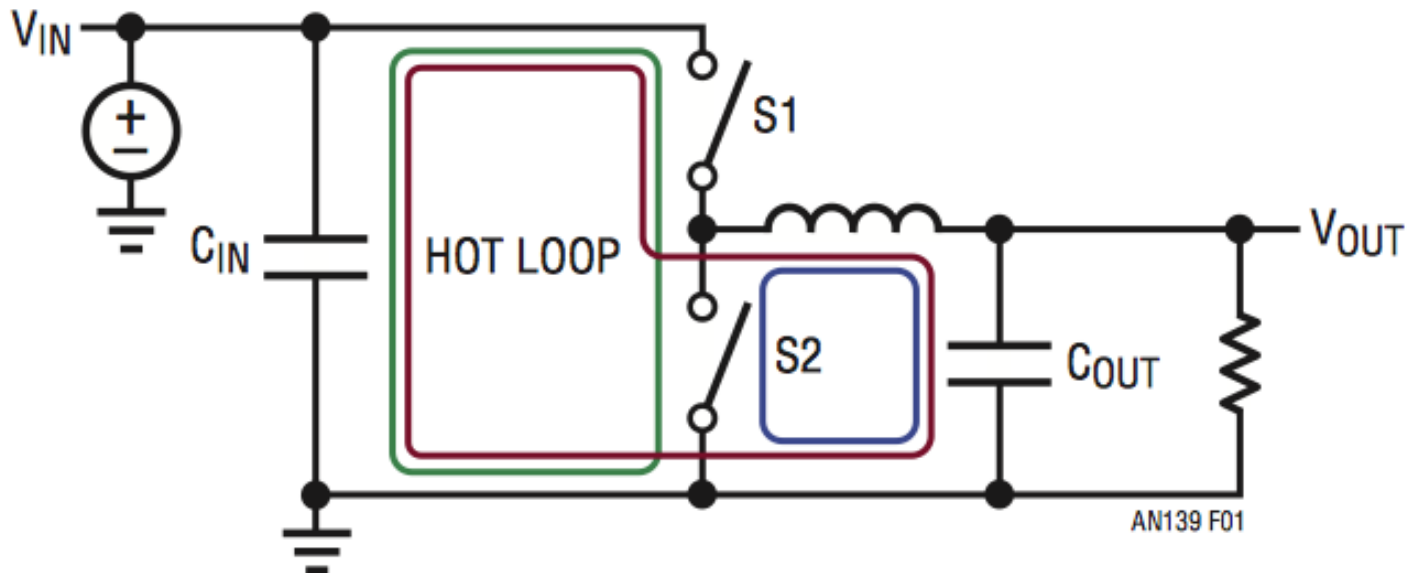


Return current

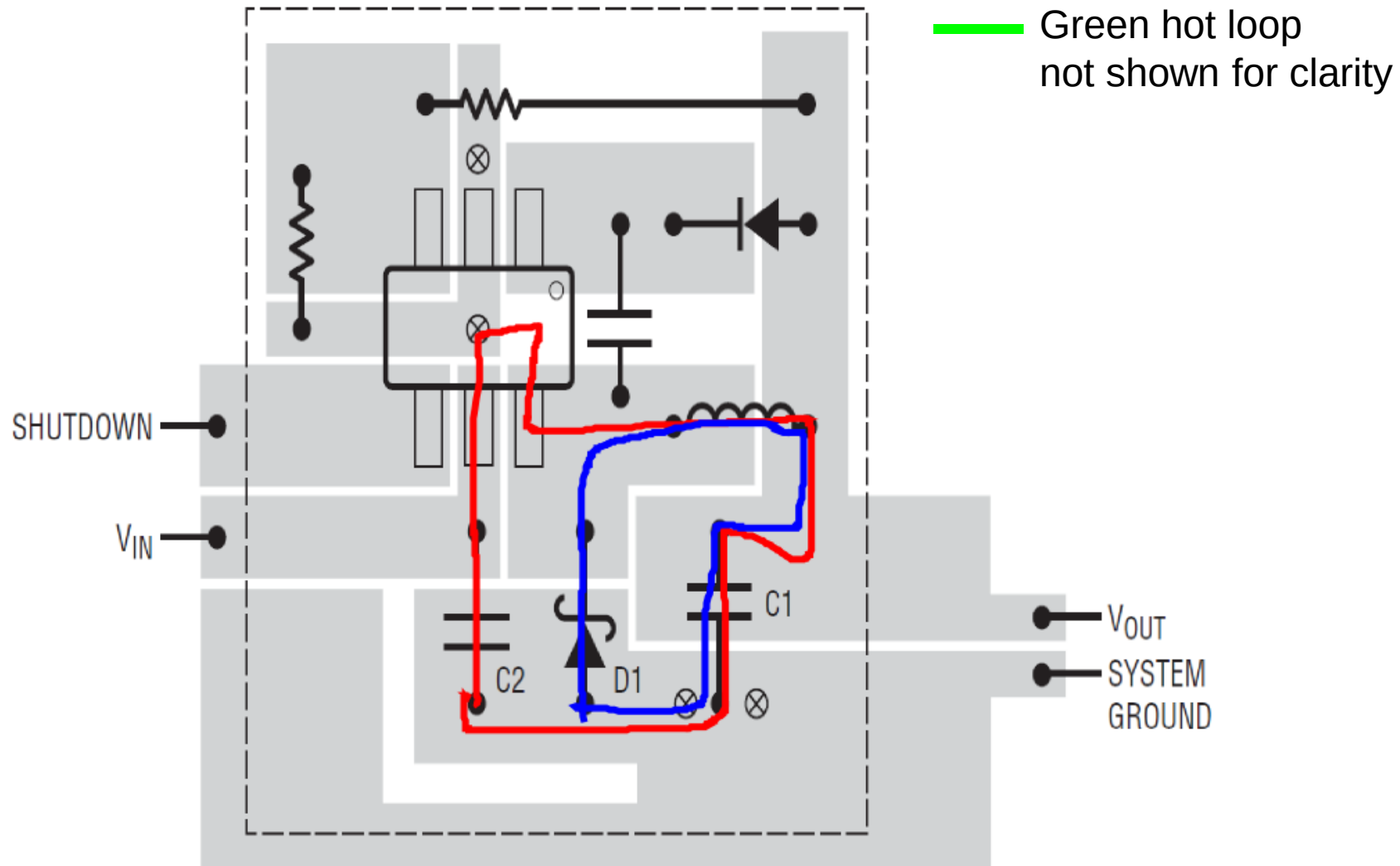


Identifying the buck hot loop

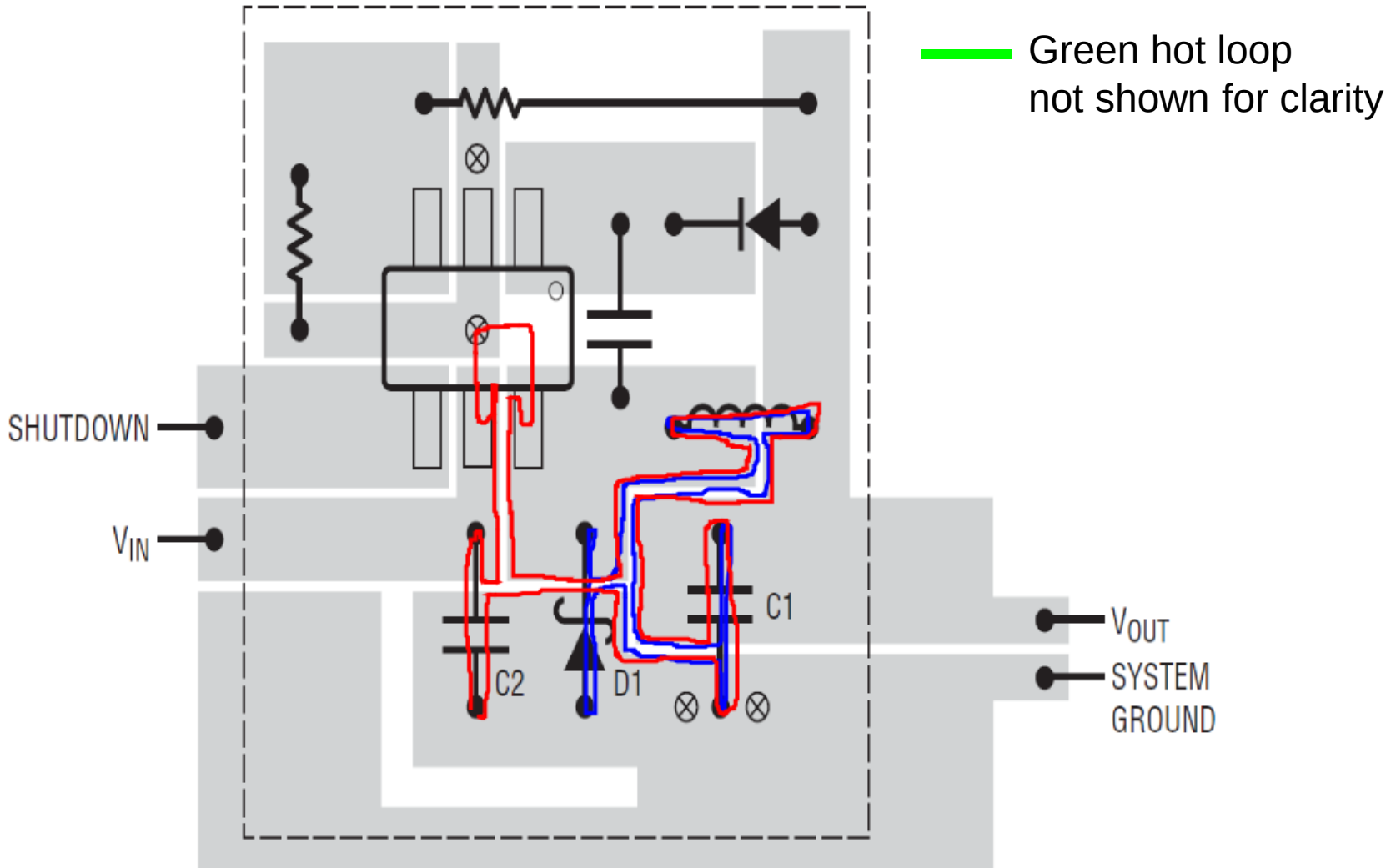
- switch on
- switch off
- Hot loop area =
area switch on - area switch off



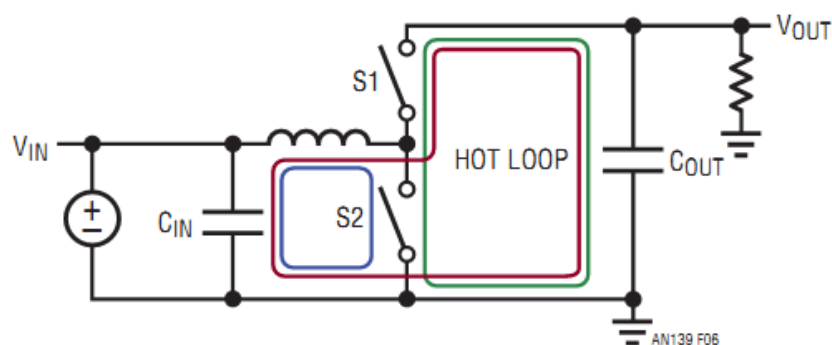
Buck hot loop < skin frequency



Hot loop > skin frequency



Hot loop examples



Boost

Figure 6

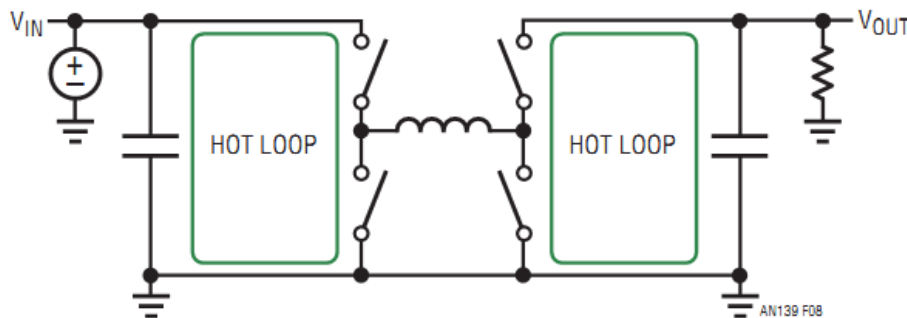


Figure 8. 4-Switch Buck-Boost

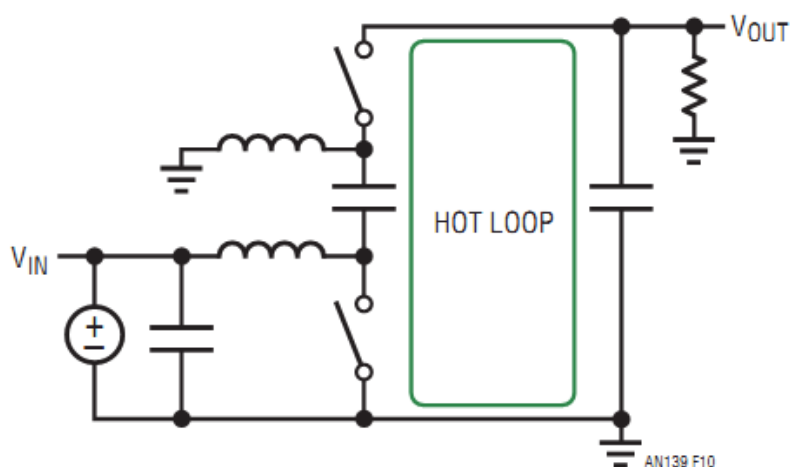


Figure 10. SEPIC

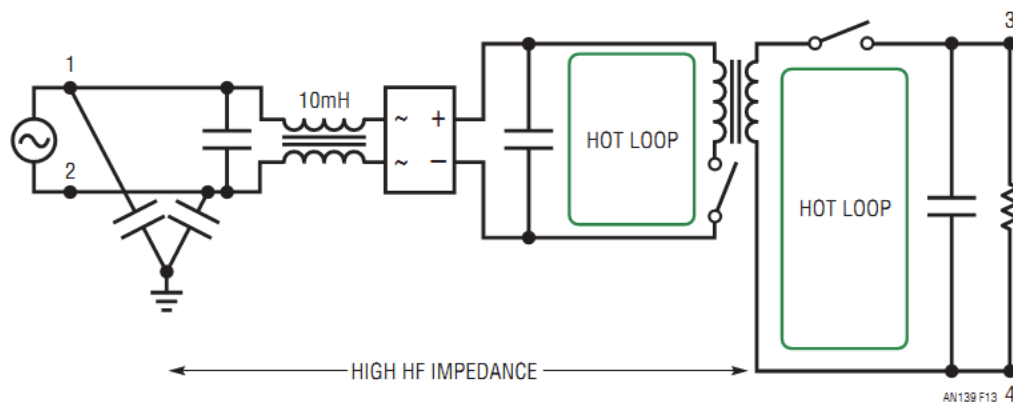


Figure 13. Isolated Offline Flyback

Other EMI sources

- Internal Vcc decoupling
 - Main switch driver
- External Switch Driver Signals

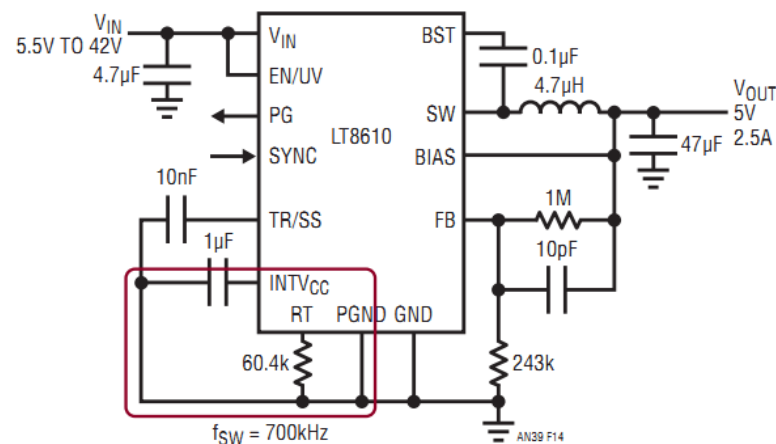


Figure 14

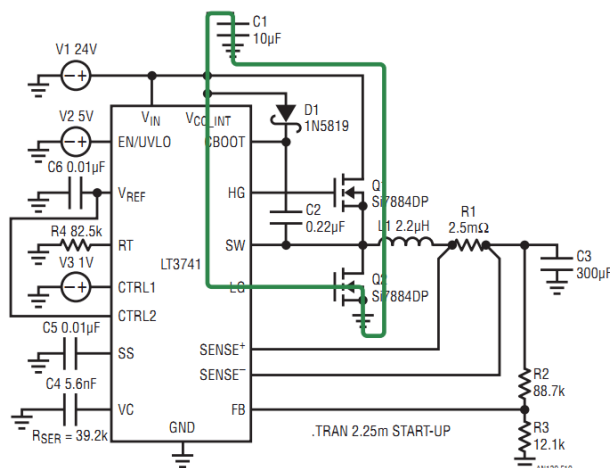


Figure 19

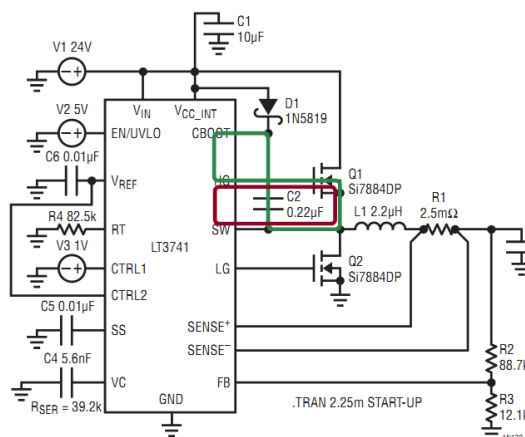


Figure 20

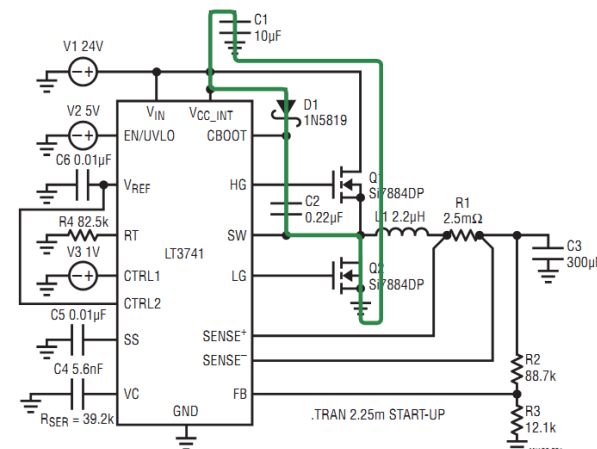
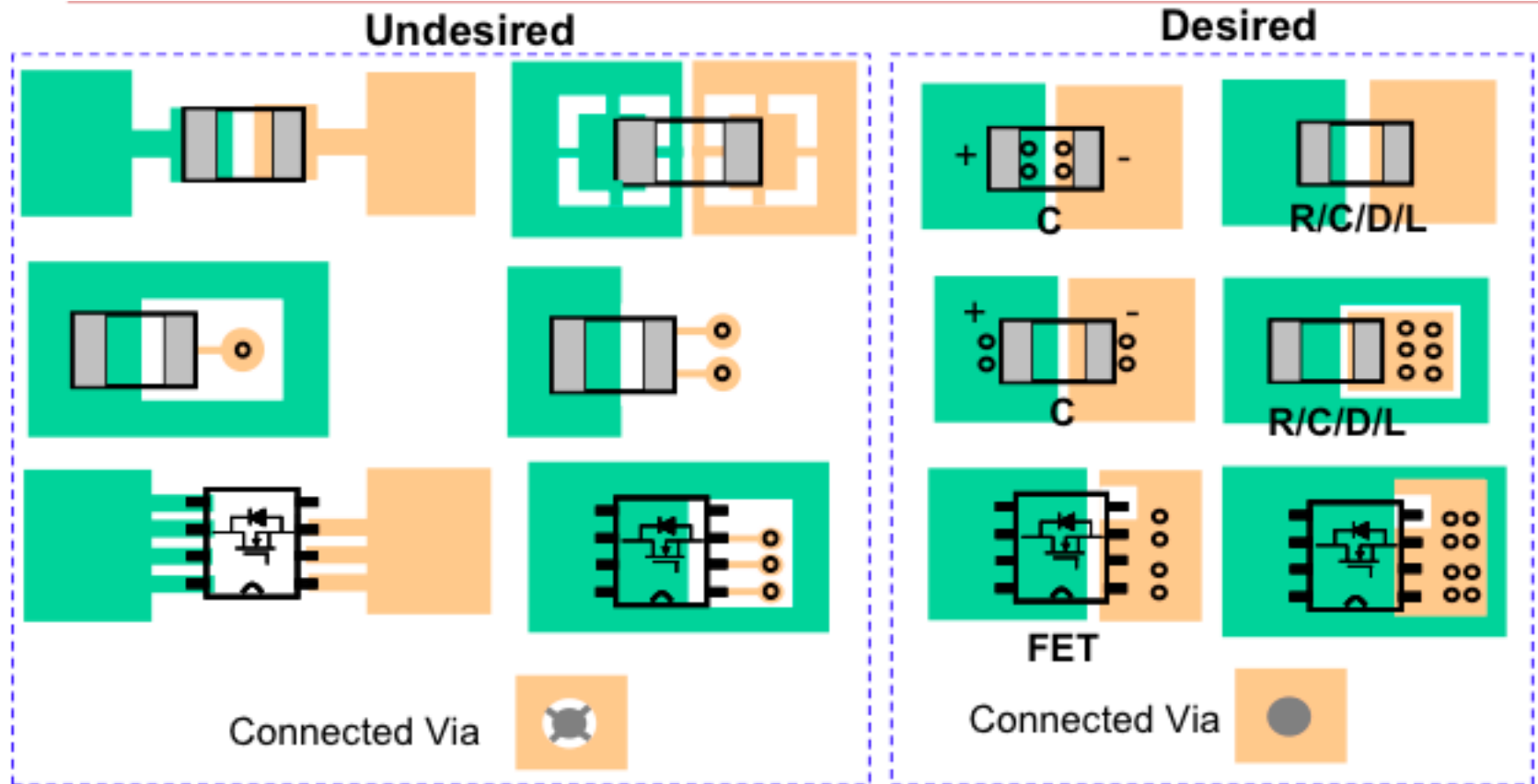
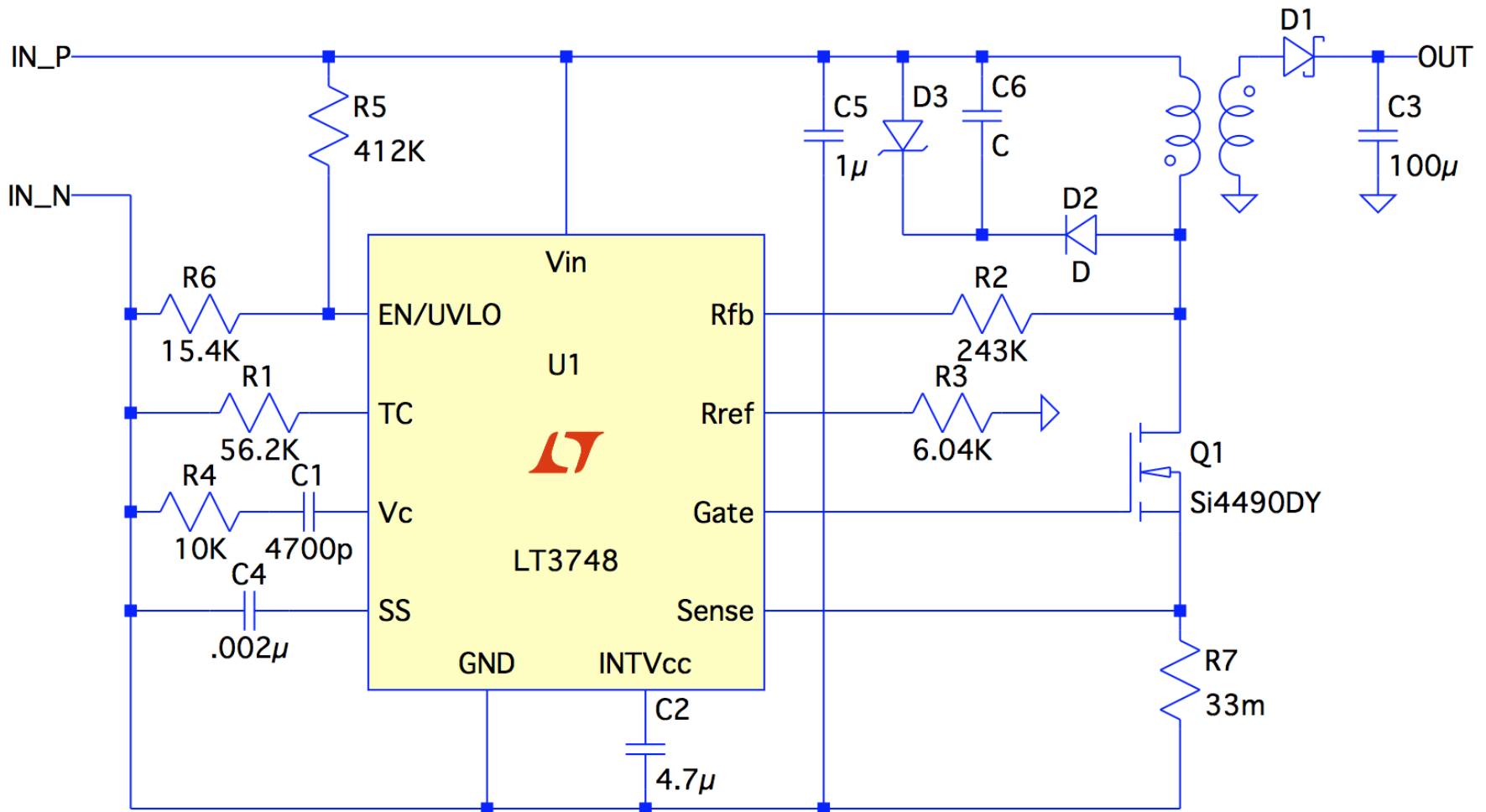


Figure 21

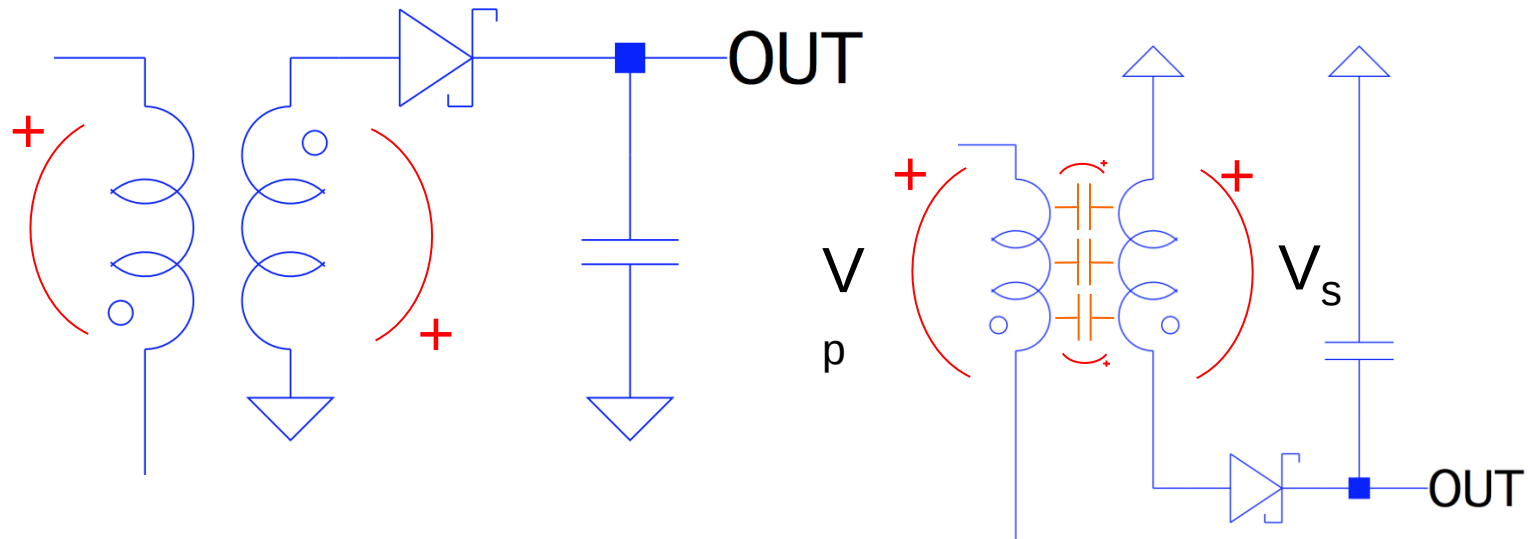
Power component footprints



Isolated converter



Transformer common mode EMI



If transformer ratio is 1:1

Then $V_p = V_s$

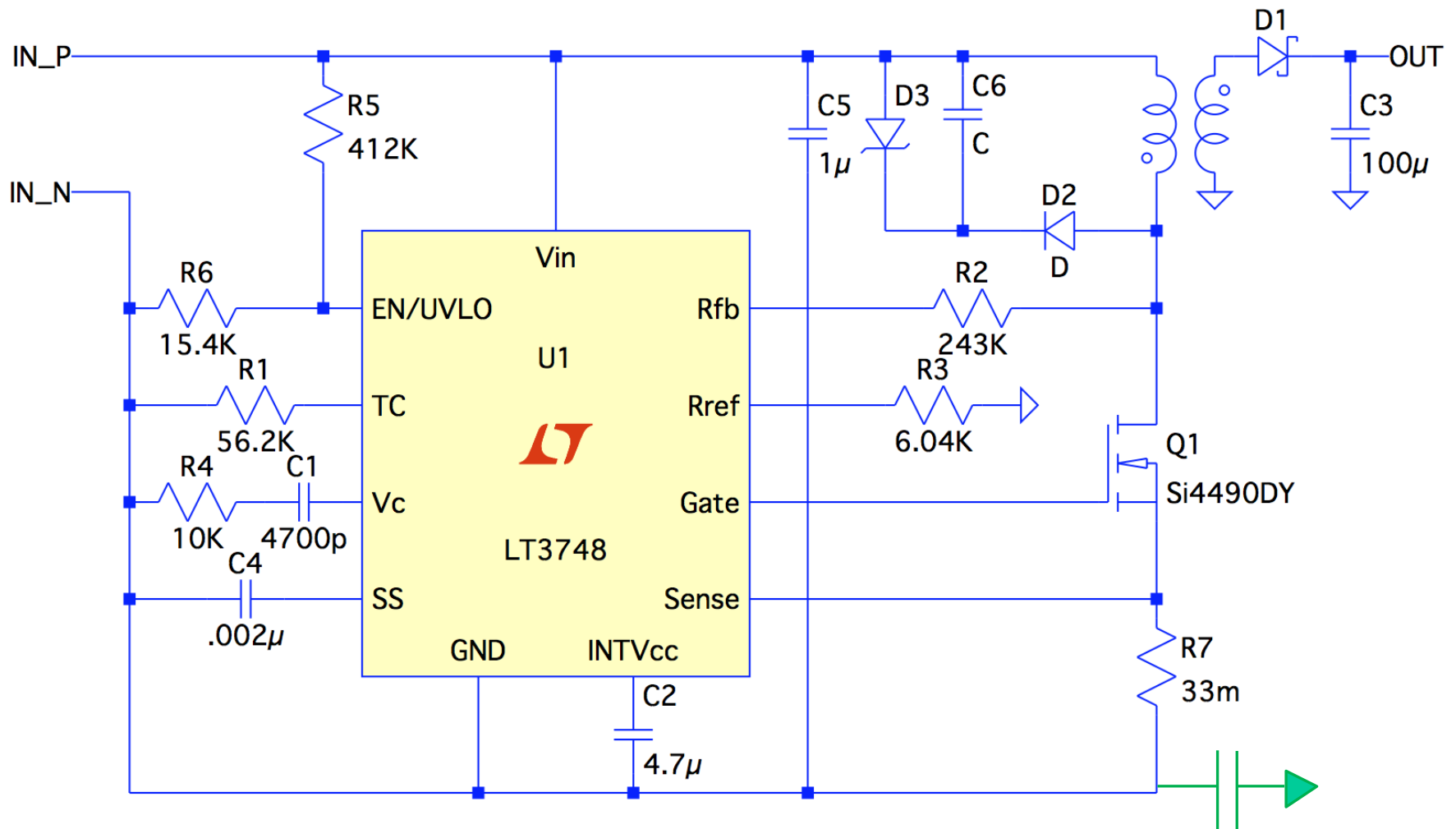
If V_p changes, V_s changes with the same voltage, Voltage between any primary to secondary winding will be constant

BUT

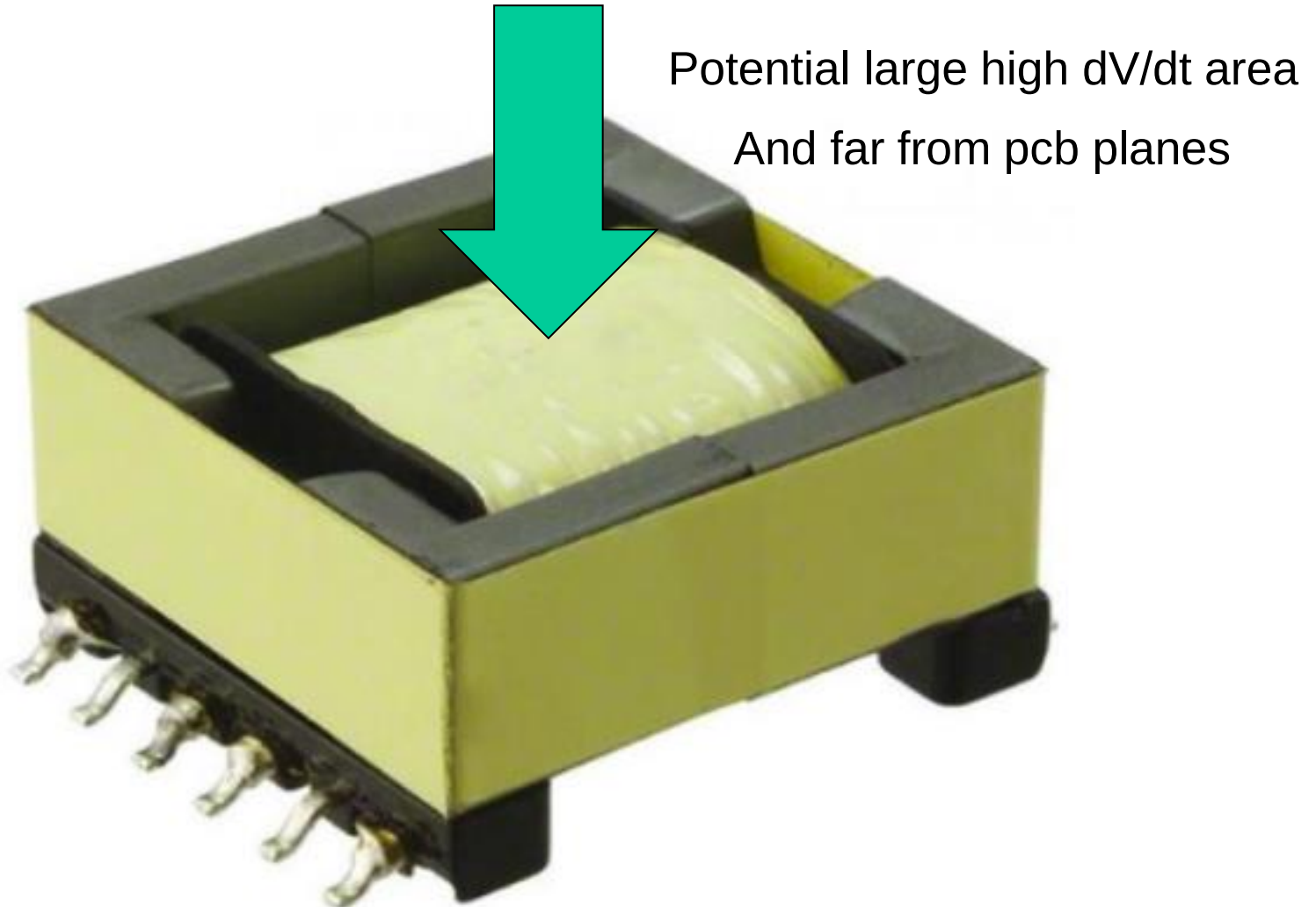
If transformer ratio is **not** 1:1

Voltage between primary to secondary windings will vary when total winding voltage changes.

THE MAGIC CAPACITOR



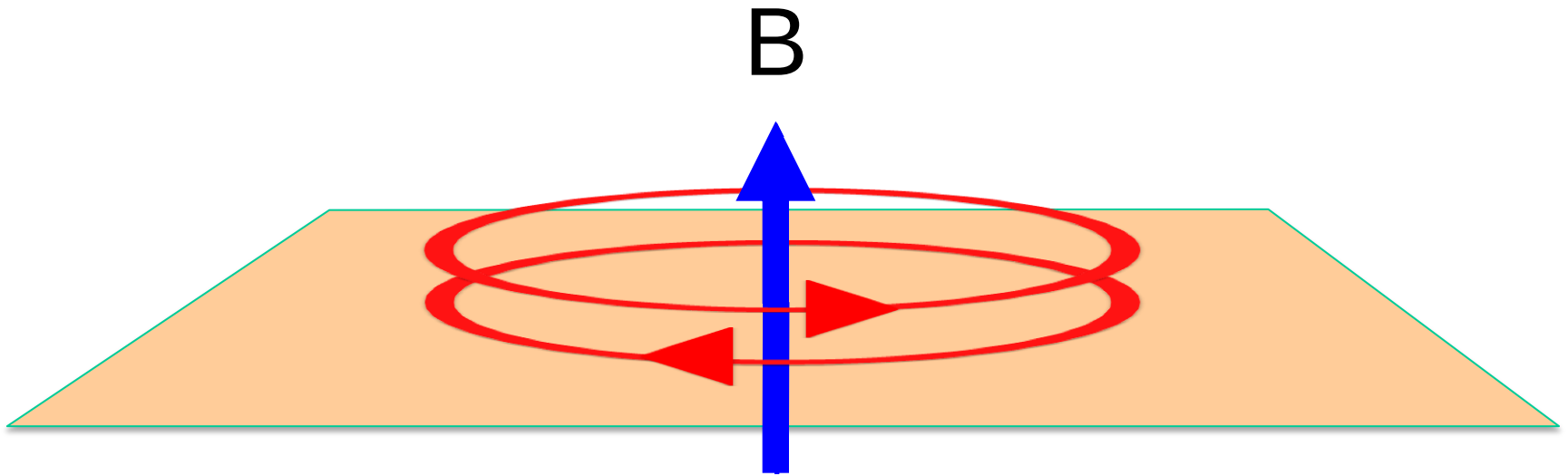
Another isolated converter issue



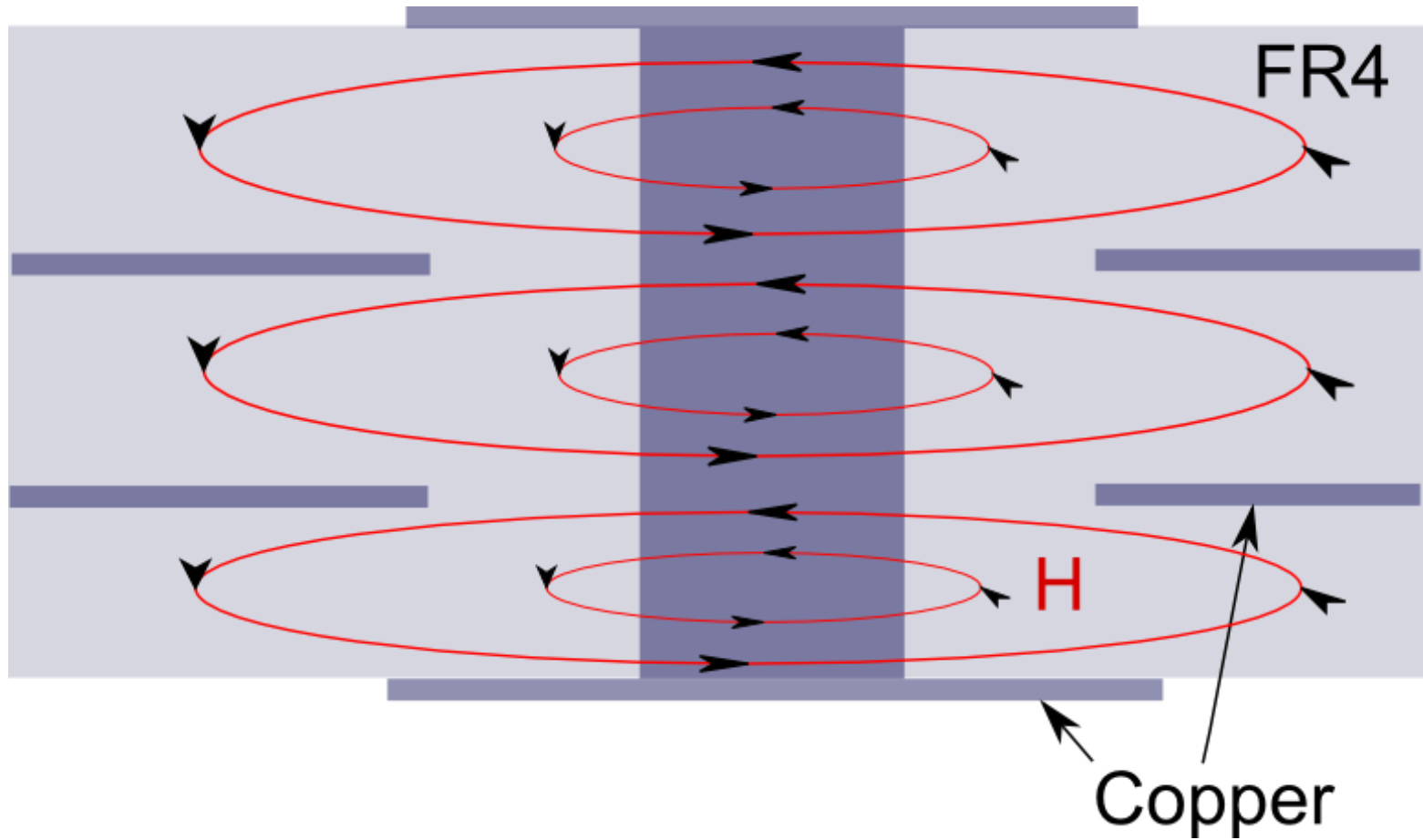
It is an antenna rising above the pcb



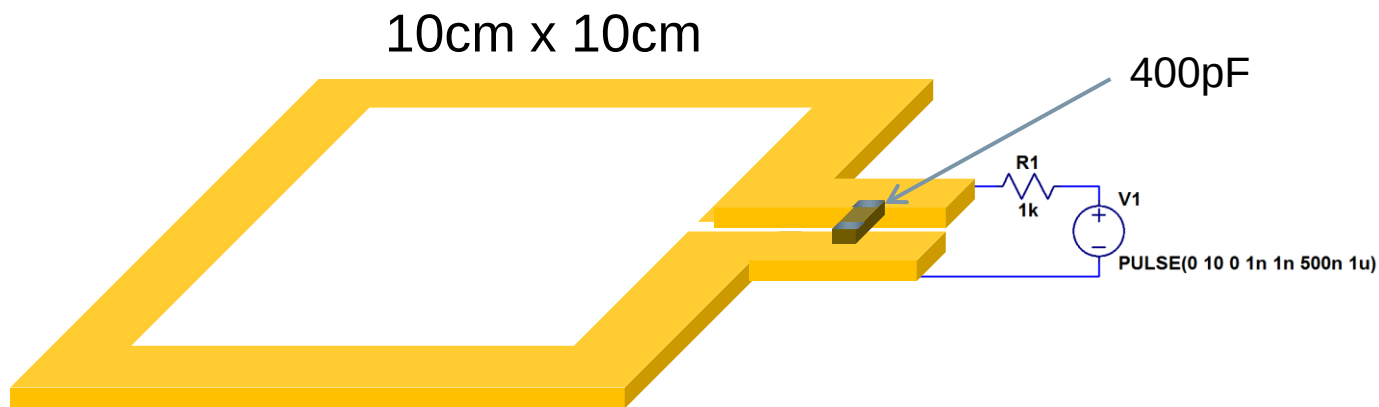
Hotloop on plane



Vias

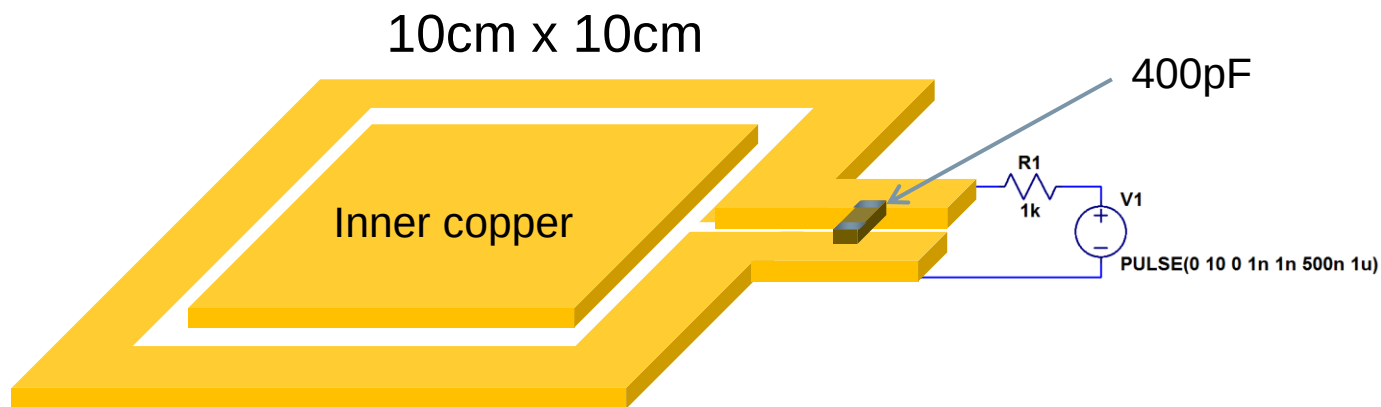


Loop short-circuit effect experiment



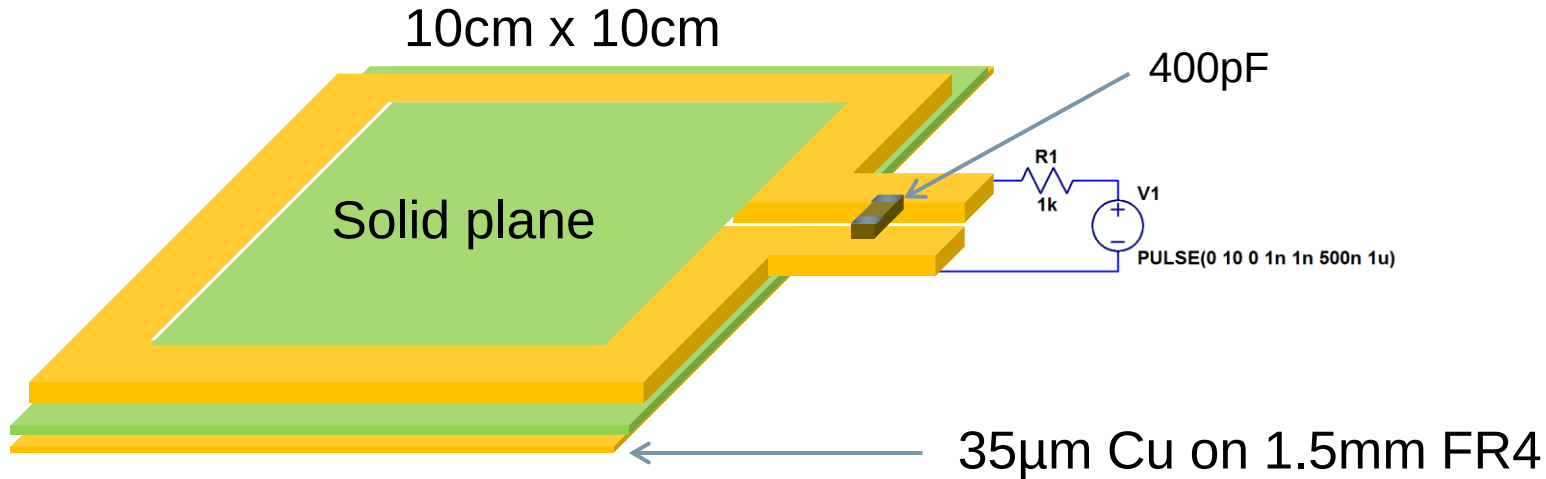
d (mm)	$f_{\text{resonance}}$ (MHz)	C (pF)	L (nH)	
n/a	18.4	400	187	Single-layer open loop

Loop short-circuit effect experiment



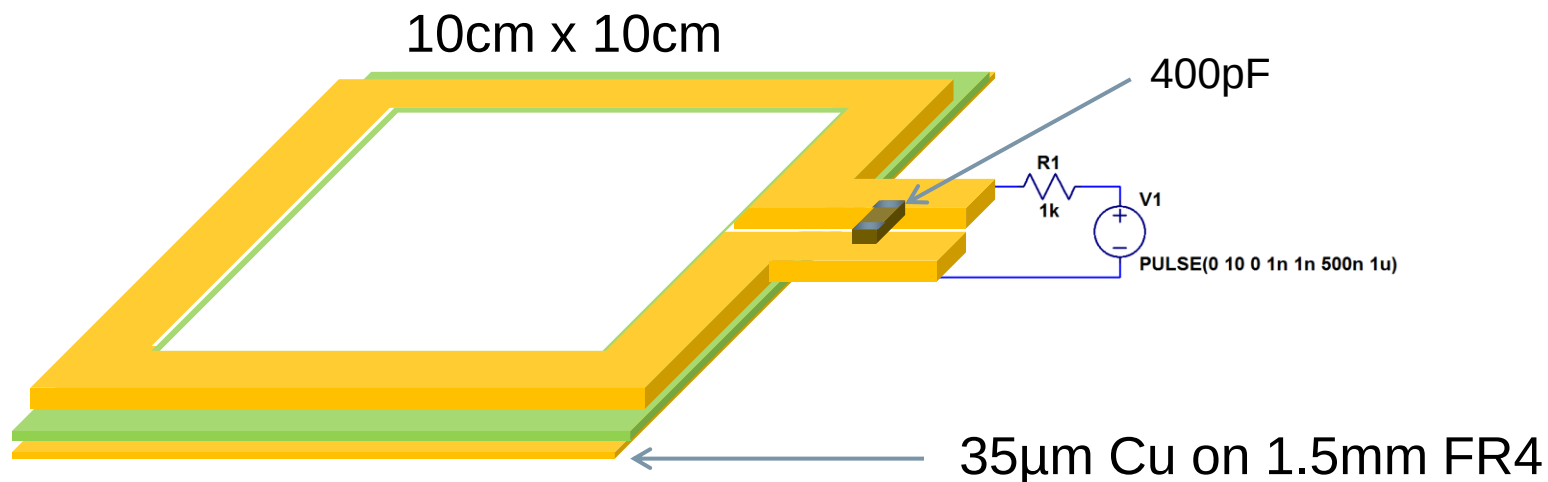
d (mm)	$f_{\text{resonance}}$ (MHz)	C (pF)	L (nH)	
n/a	18.4	400	187	Single-layer open loop
n/a	21.2	400	141	Inner copper Short-Circuit loop

Loop short-circuit effect experiment



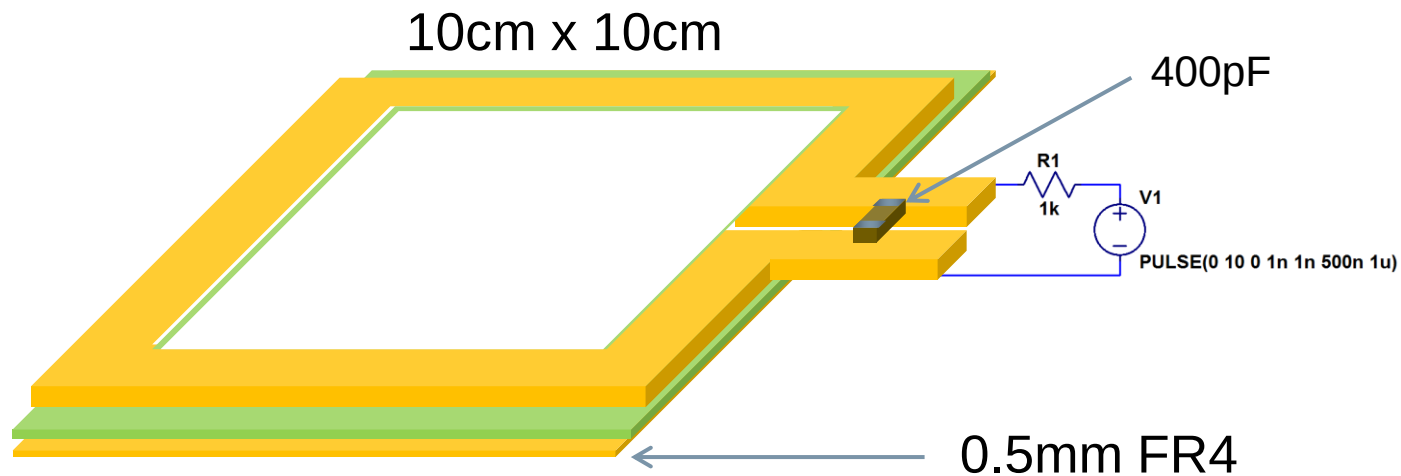
d (mm)	f _{resonance} (MHz)	C (pF)	L (nH)	
n/a	18.4	400	187	Single-layer open loop
n/a	21.2	400	141	Inner copper Short-Circuit loop
1.5	38.9	400	42	Solid Plane

Loop short-circuit effect experiment



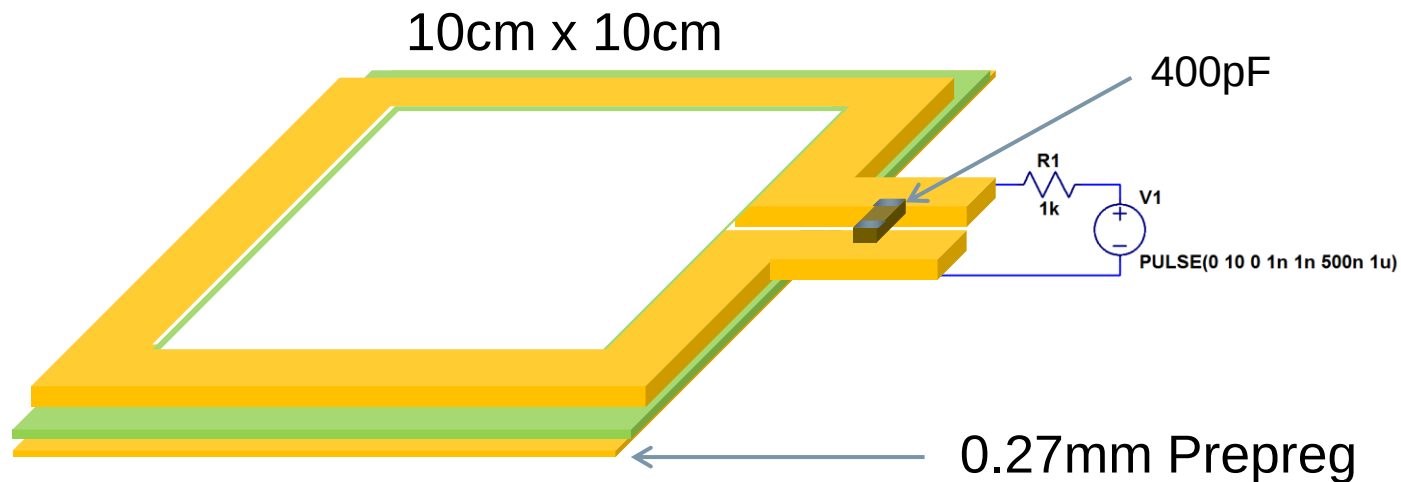
d (mm)	f _{resonance} (MHz)	C (pF)	L (nH)	
n/a	18.4	400	187	Single-layer open loop
n/a	21.2	400	141	Inner copper Short-Circuit loop
1.5	38.9	400	42	Solid Plane
1.5	34.7	400	53	Rectangular Short-Circuit loop plane

Loop short-circuit effect experiment



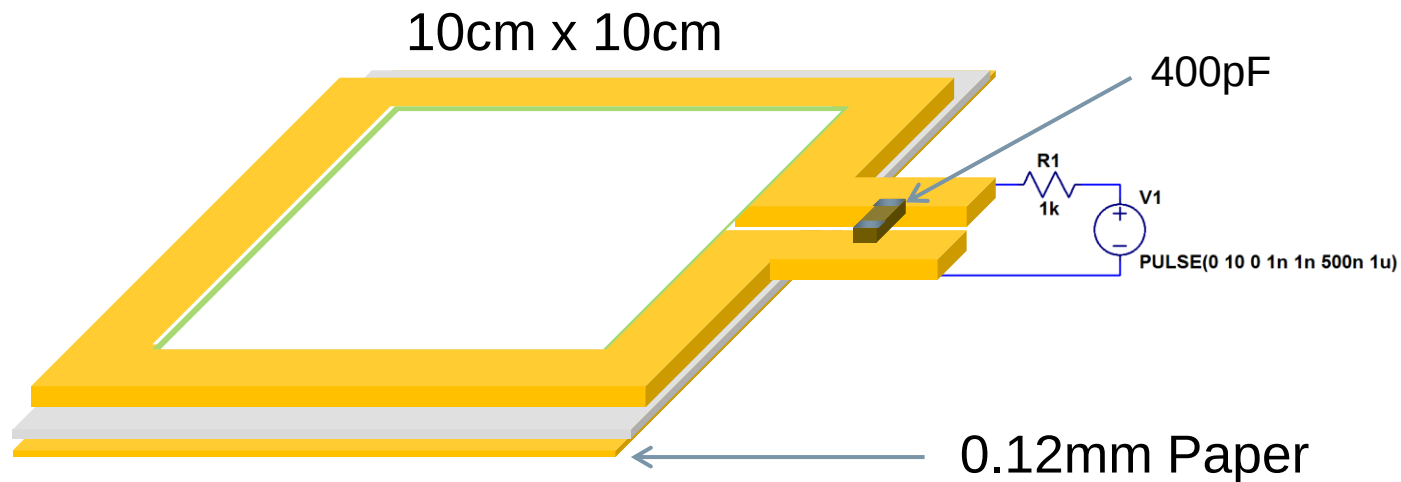
d (mm)	$f_{\text{resonance}}$ (MHz)	C (pF)	L (nH)	
n/a	18.4	400	187	Single-layer open loop
n/a	21.2	400	141	Inner copper Short-Circuit loop
1.5	38.9	400	42	Solid Plane
1.5	34.7	400	53	Rectangular Short-Circuit loop plane
0.5	52.1	400	23	Thin Rectangular

Loop short-circuit effect experiment



d (mm)	f _{resonance} (MHz)	C (pF)	L (nH)	
n/a	18.4	400	187	Single-layer open loop
n/a	21.2	400	141	Inner copper Short-Circuit loop
1.5	38.9	400	42	Solid Plane
1.5	34.7	400	53	Rectangular Short-Circuit loop plane
0.5	52.1	400	23	Thin Rectangular
0.27	55	400	21	

Loop short-circuit effect experiment



d (mm)	$f_{\text{resonance}}$ (MHz)	C (pF)	L (nH)	
n/a	18.4	400	187	Single-layer open loop
n/a	21.2	400	141	Inner copper Short-Circuit loop
1.5	38.9	400	42	Solid Plane
1.5	34.7	400	53	Rectangular Short-Circuit loop plane
0.5	52.1	400	23	Thin Rectangular
0.27	55	400	21	
0.12	69	400	13	Paper

Loop short-circuit effect experiment

d (mm)	f _{resonance} (MHz)	C (pF)	L (nH)	
n/a	18.4	400	187	Single-layer open loop
n/a	21.2	400	141	Inner copper Short-Circuit loop
1.5	38.9	400	42	Solid Plane
1.5	34.7	400	53	Rectangular Short-Circuit loop plane
0.5	52.1	400	23	Thin Rectangular
0.27	55	400	21	
0.12	69	400	13	Paper

- Inductance goes down with distance to next layer
- Multilayer PCB will have less inductance than 2-layer 1.5mm
- Thinner = better
- Doesn't need to be a GND-plane

Power Supply Layout and EMI

Christian Kueck

Introduction

PC-board layout determines the success or failure of every power supply project. It sets functional, electromagnetic interference (EMI), and thermal behavior. Switching power supply layout is not black magic, but is often overlooked until it is too late in the design process. Fortunately physics is on your side. Functional and EMI requirements must be met, and in a world of trade-offs in power supply unit layout, what is good for functional stability is good for EMI. Good layout from first prototyping on does not add to cost, but actually saves significant resources in EMI filters, mechanical shielding, EMI test time and PC board runs. This application note focuses primarily on nonisolated topologies, but will examine some isolated topologies as well. You will learn to make the optimum choices regarding PC-board layout for solid power supply designs.

I remember about a dozen years ago as a customer was

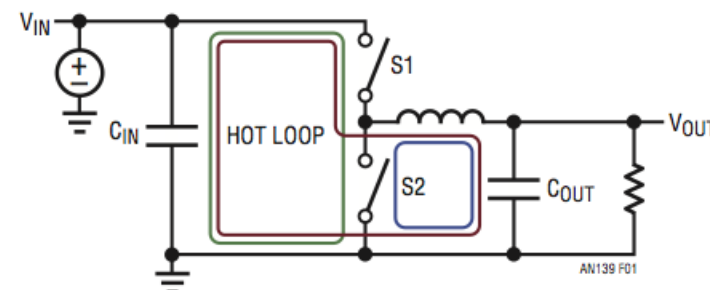


Figure 1

During the on cycle with S1 closed and S2 open, the AC current follows the red loop (Figure 1). During the off cycle, with S1 open and S2 closed, the AC current follows the blue loop. Both currents have a trapeze shape. People often have difficulty grasping that the loop producing the highest EMI is not the red nor the blue loop. Only in the green loop flows a fully switched AC current, switched from zero to I_{PEAK} and back to zero. We refer to the green loop as a hot loop, since it has the highest AC and EMI energy.

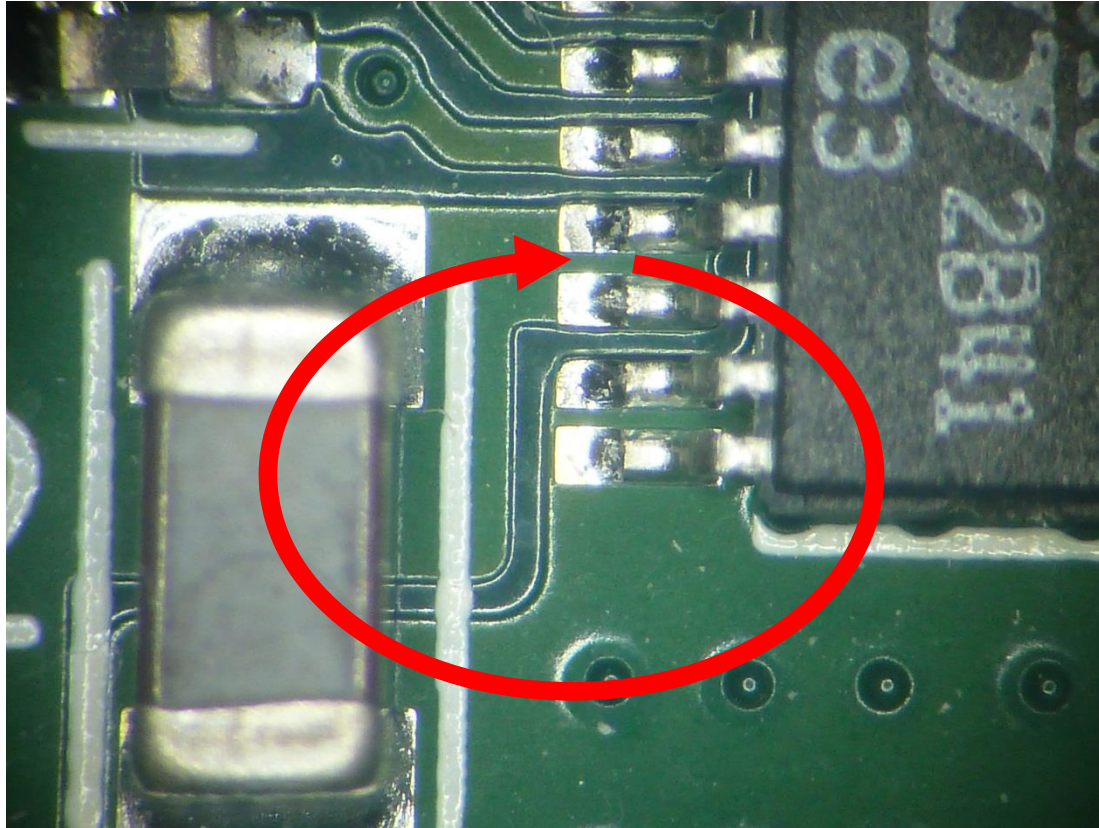
SILENCE PLEASE

Linear technology invented
noise cancelling
for switch mode regulators.

It is called Silent Switcher¹

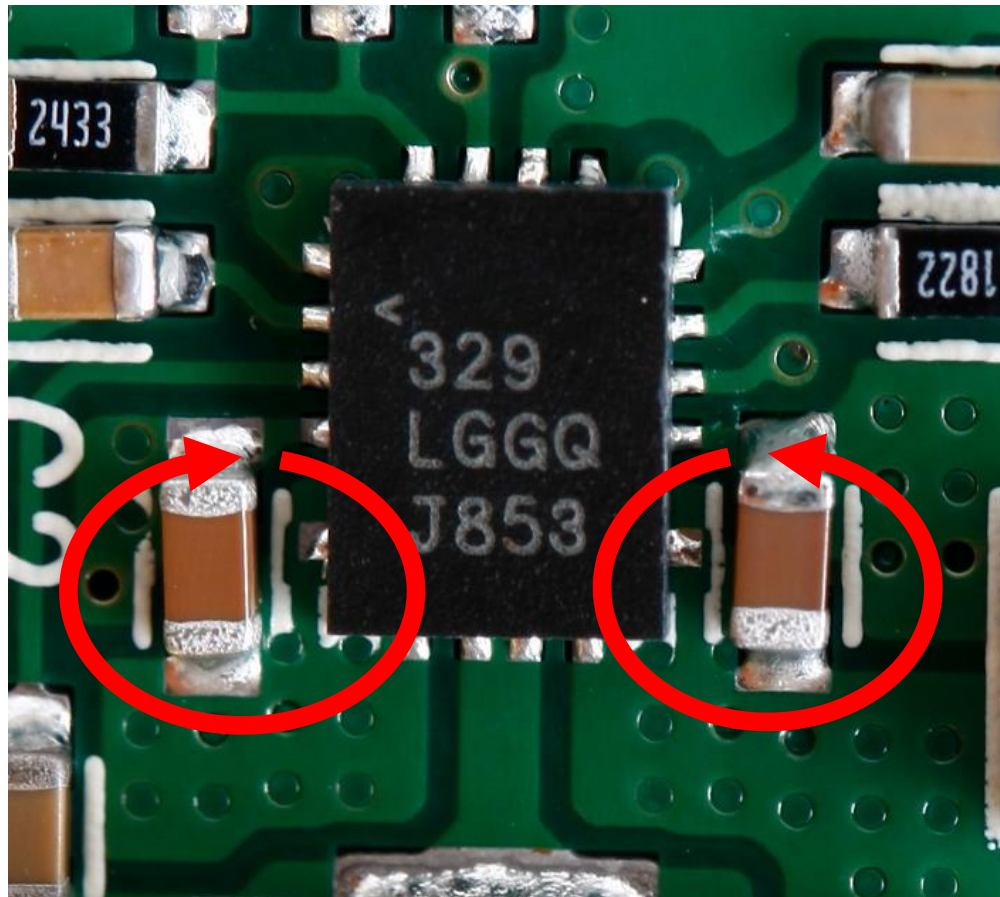
(1) Silent Switcher is a registered trade mark and Patent pending from Linear Technology Corp.

How does it work



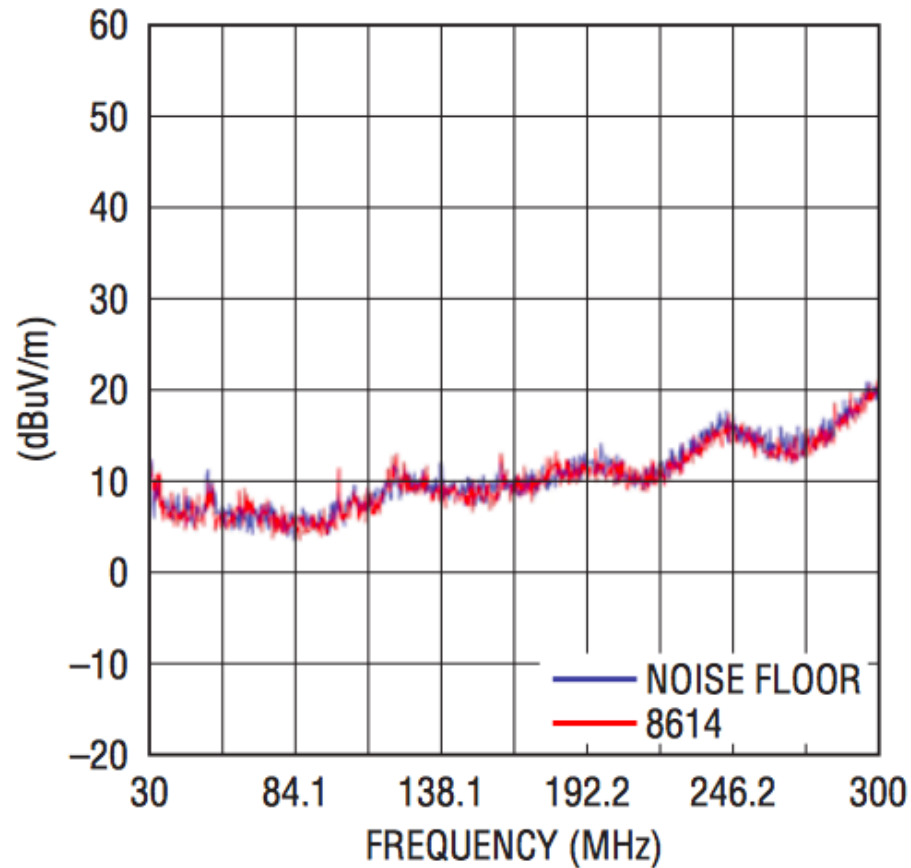
Even on the most compact design the decoupling components with the switch have some finite dimensions and create an EMI field.

LT8614 Silent Switcher



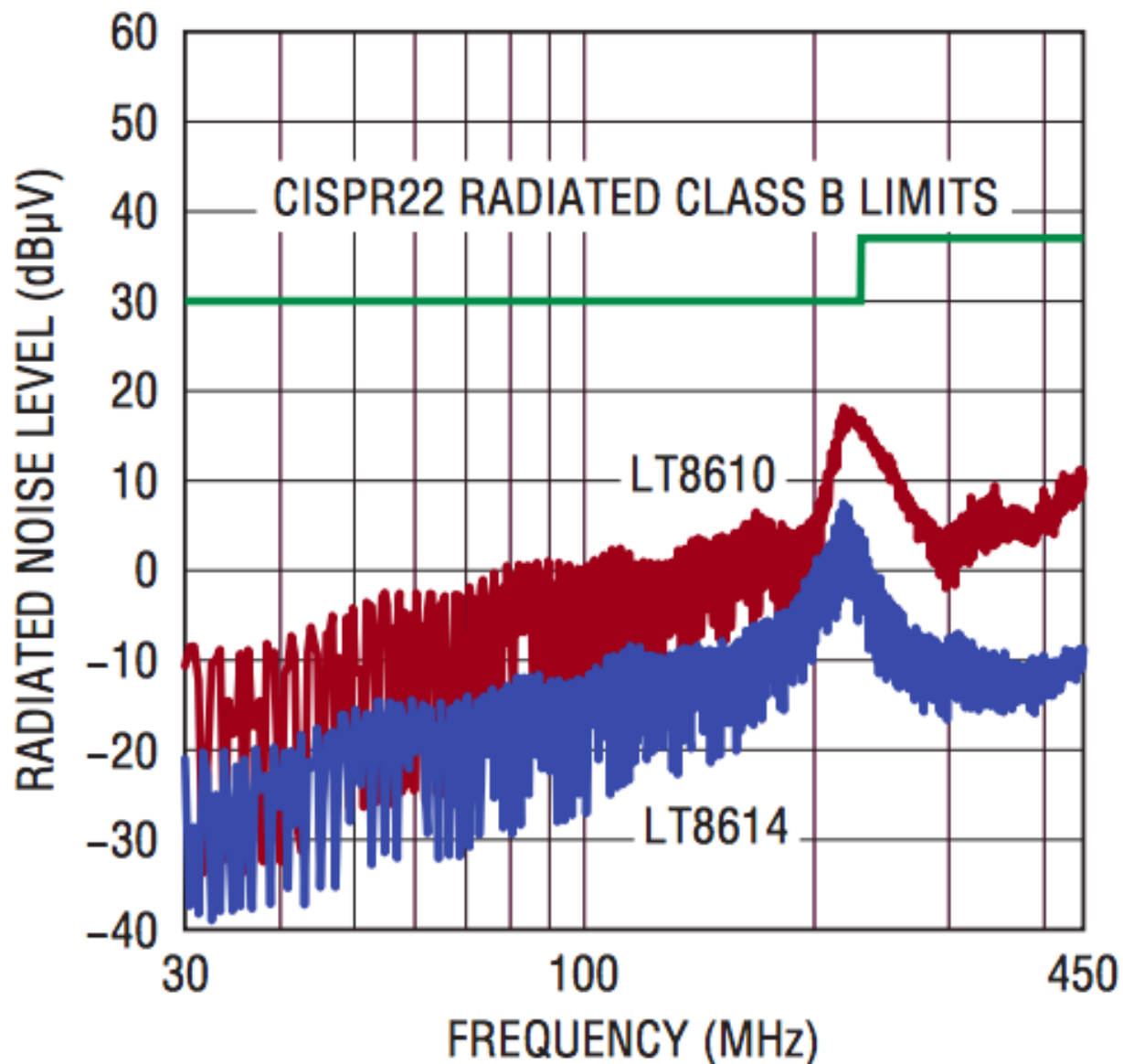
On a Linear Technology Silent Switcher two identical decoupling loops with opposite orientation cancel each others EMI field.

LT8614 EMI Performance



an144 F01

Figure 1. Noise Floor and the LT8614 Board at CISPR25 Radiated Measurement in an Anechoic Chamber



an144 F02

**Figure 2. LT8610 and LT8614 700kHz 14V to 3.3V 2A
Radiated EMI in GTEM Corrected for OATS**



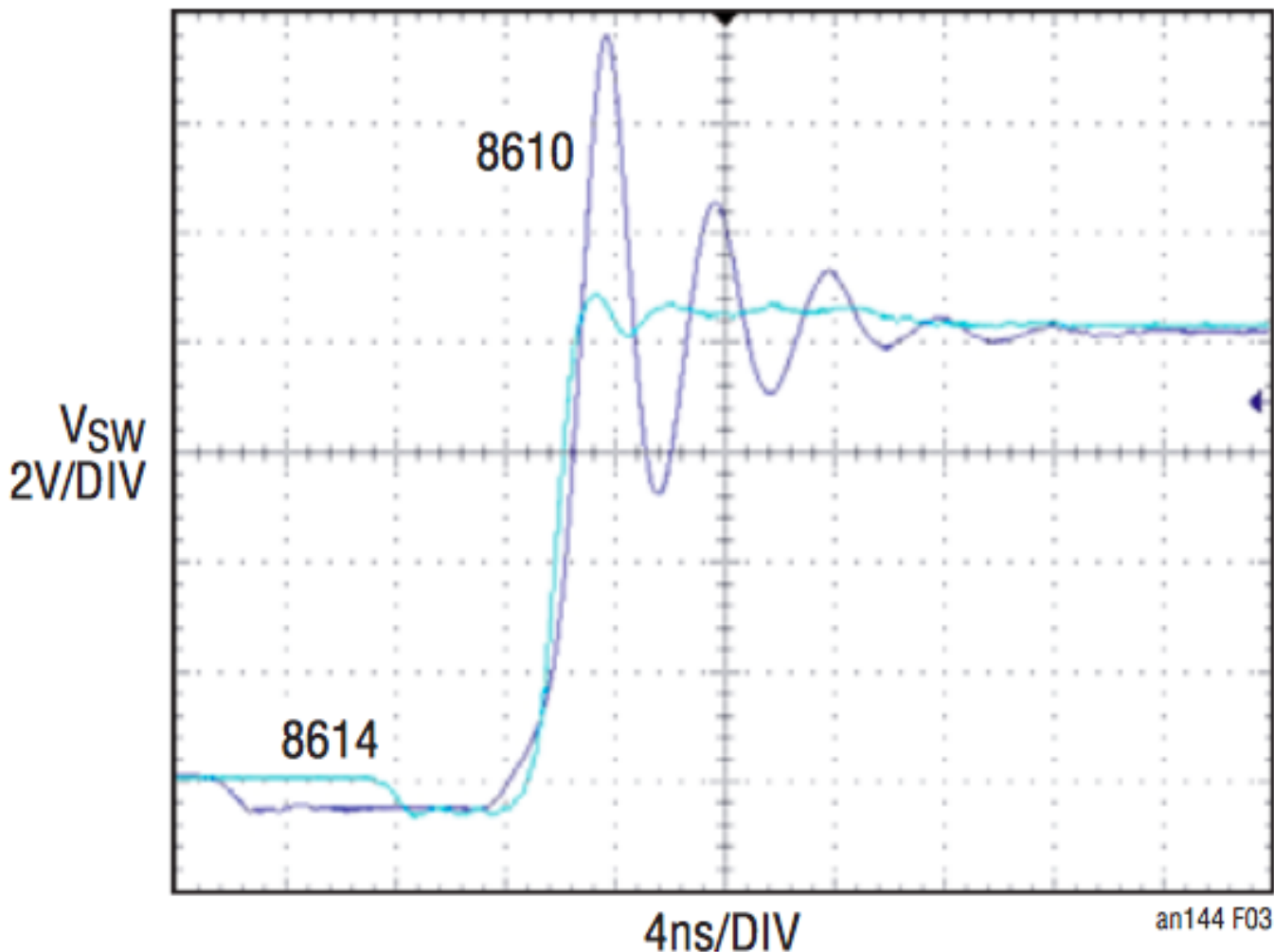
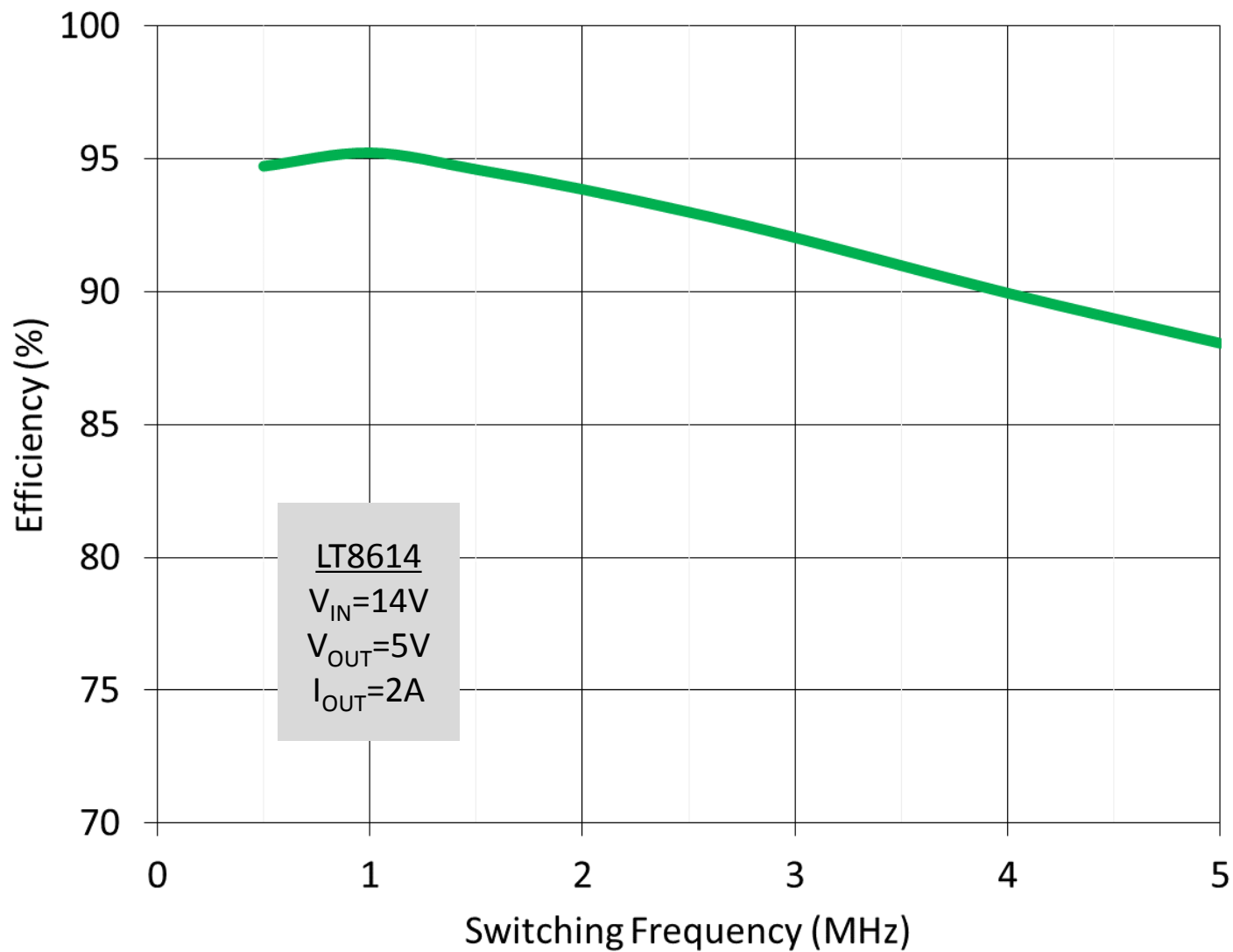


Figure 3. LT8610 and the LT8614, Switch Node Rising Edge Both at $8.4V_{IN}$, $3.3V_{OUT}$ at 2.2A

Efficiency vs. Switching Frequency



–LT8614–



Application Note 144

November 2013

Reduce EMI and Improve Efficiency with Silent Switcher Designs

Christian Kueck

Questions ?

Thank you